Aalto University
School of Science
Degree Programme in Computer Science and Engineering

Kirti Prakash

A binary combinatorial histone code

Master's Thesis

Espoo, February 29, 2012

Supervisor and Instructor: Professor Harri Lähdesmäki

AALTO UNIVERSITY
SCHOOL OF SCIENCE

ABSTRACT OF THE
MASTER'S THESIS

Author:	Kirti Prakash
Name of the Thesis:	A binary combinatorial histone code
Date:	February 29, 2012
Number of pages:	x + 59
Department:	Department of Information and Computer Science
Professorship:	T-61 Information and Computer Science
Supervisor and Instructor :	Professor Harri Lähdesmäki

Background:
Post-translational modifications of nucleosomal histone proteins play a vital role in regulation of gene expression, and is considered as one of the major emerging concept to study gene regulation. But a comprehensive study of combinatorial patterns, or networks, of the post-translational histone modifications within nucleosomes still remains limited.

Methods and Results:
In this thesis, we describe a framework to identify histone modification networks that might exist within nucleosomes. To test this framework, we combined ChIP-seq data from human CD4+ T cells covering 39 different histone modifications, and identified over 750,000 nucleosomes. We then carried out boolean association analysis for top 186,000 nucleosomes for each individual histone modification, and robustly searched to find which other modifications show best dependency. We then identified most prominent histone modification combination that occurred among the selected modifications.

Conclusions:
Our results provide a comprehensive list of combinatorial patterns of histone modifications that might exist together within nucleosomes. We conclude that Boolean approach provides a computationally efficient way to study histone modification networks. We believe that when this approach is extended to different regulatory regions, such as enhancers and insulators, will provide us with better insights into gene regulation.

Keywords: histone modifications, nucleosomes, chromatin, gene regulation, regulatory elements, ChIP-Seq data, boolean models, feature selection, network selection

Acknowledgments

First, I would like to thank my promoter Harri Lähdesmäki, whose both active and poised nature were most helpful for facilitating my transcription. I would also like to thank my proximal promoter Antti Larjo for increasing my transcription level.

Next I would like to thank my enhancers Fabian Hoti and Erik Aurell, for relaxing me at various regulatory sites through their complex HAT activity. I would also like to thank my distal enhancers Sirkka-Liisa Varvio and Veli Mäkinen, who were most encouraging when I began my transcription in Finland. I would also like to thank my mediators Satu Kähäri and Juha Jokinen for making my life easy with administrative and technical matters.

Next, I would like to thank my insulator mom and LCR brother. They were always true to their biological representation. I would also like to thank my activators Shekhar, Ajay, Elia and Maria. I also thank all my silencers and repressors, who I think are really important to have.

Finally, I thank my ATCG dad, who has always supported me, unconditionally. I dedicate this work to him.

Apart from courtesy, this acknowledgement also provides a brief introduction to things to follow. Indeed it was a concerted work of various regulatory elements to direct my transcription. Thank you all.

Kirti Prakash
(Gene, mostly introns, but highly acetylated)

Contents

Abstract **ii**

Acknowledgments **iii**

Abbreviations **v**

List of Figures **viii**

Preface **ix**

1 Epigenetics and histone modifications **1**

1.1 Chromatin 2

1.2 Post-translational modifications on histones 5

1.3 The Histone cross talk 10

2 Histone modification data processing **12**

2.1 ChIP-Sequencing 12

2.2 Mapping short reads 14

2.3 Identifying the nucleosome positions 16

2.4 The binary histone modification data 17

2.5 The correlation plot of histone modifications 18

2.6 Hierarchical clustering of histone modifications 20

3 Gene regulation and profiles of histone modifications at various regulatory regions **21**

3.1 Transcriptional machinery and histone modifications 22

3.2 Histone modifications at predicted nucleosomes 23

3.3 Histone modifications at TSS 27
3.4 Histone modifications at RNA Pol II 29
3.5 Histone modifications at enhancers 33
3.6 Histone modifications at CTCF 35

4 Learning combinatorial patterns of histone modifications under Boolean network model 41
4.1 Background and definitions 42
4.2 Best fit algorithm 42
4.3 Hold-out cross validation 44
4.4 Selecting histone modifications 44
4.5 Selecting histone modification networks 46
4.6 Results 47

5 Summary and future work 52
5.1 Summary and conclusions 52
5.2 Future work 53

Abbreviations

DNA	Deoxyribonucleic acid
PTMs	Post Translational Modifications
K	Lysine
R	Arginine
S	Serine
Y	Tyrosine
Ac	Acetylation
Me	Methylation
P	Phosphorylation
Ub	Ubiquitylation
Su	Sumoylation
HATs	Histone Acetyl Transferases
HDACs	Histone DeACetylases
TSS	Transcription Start Site
LCR	Locus Control Region
H3K4me1	Histone 3 Lysine 4 mono-methylation
H3K4me2	Histone 3 Lysine 4 di-methylation
H3K4me3	Histone 3 Lysine 4 tri-methylation
H3K27ac	Histone 3 Lysine 27 acetylation

List of Figures

1.1 The different levels of chromatin structure 2
1.2 Schematic representation of formation of histone octamer . . . 4
1.3 The representation of chromatin 5
1.4 Various post-translational modifications on Histone tails . . . 7
1.5 Acetylation and deacetylation of Lysines 8
1.6 Varying degree of methylations on Lysine residue 9
1.7 Histone cross talk 11

2.1 Chromatin immunoprecipitation steps 13
2.2 Burrows-Wheeler transform approach to map reads 14
2.3 Identification of nucleosomes from ChIP-Seq data 16
2.4 Assignment of histone modifications to Nucleosomes 17
2.5 The correlation plot of histone modifications 19
2.6 The hierarchical clustering of histone modifications 20

3.1 Histone modifications at various gene regulatory regions 22
3.2 Histone modifications at Nucleosomes: Part 1 24
3.3 Histone modifications at Nucleosomes: Part 2 25
3.4 Histone modifications at TSS: Part 1 28
3.5 Histone modifications at TSS: Part 2 30
3.6 Histone modifications at RNAPol2: Part 1 31
3.7 Histone modifications at RNAPol2: Part 2 33
3.8 Histone modifications at Enhancers: Part 1 36
3.9 Histone modifications at Enhancers: Part 2 37
3.10 Histone modifications at CTCF: Part 1 38
3.11 Histone modifications at CTCF: Part 2 40

5.1 The overall workflow of thesis 53

Preface

> *When Adam tries to find a help-mate, none of the animals are satisfactory, and so God causes the man to sleep, and creates a woman from his rib. The man names her "Woman" , "for this one was taken from a man". On account of this a man leaves his father and his mother and clings to his woman*
>
> — Genesis 2
>
> *Male:Female::DNA:Histone*

If we momentarily consider DNA to be the man and histones to be the woman then the "Histone-DNA" interaction is not too different from one described in Genesis 2. Histone proteins, after the process of transcription and translation, are created from DNA sequences and in-turn DNA clings to these proteins, resulting in formation of nucleosomes. Nucleosomes are basic units that are responsible for DNA coiling and formation of chromosomes. Nucleosomes, post translational modifications on histones, the interaction between these modifications and their role in gene regulation are the central subject of this thesis. In this thesis, we provide a framework starting from raw sequenced reads to finding complex networks that might exist on the tails of histone proteins.

Outline of the thesis

In chapter 1, we provide basic but sufficient biological background to understand chromatin biology, different types of modifications that occur on the tails of histone proteins, and the histone code hypothesis.
In chapter 2, we start by providing a brief description about ChIP-Seq protocol. We then describe various software's used to process ChIP-Seq data and identify nucleosome-positions from it. We also provide correlation and clustering results of the histone modification assignment to the predicted nucleosomes.
In chapter 3, we study the relationship between histone modifications and gene regulation, and provide the profiles of 39 histone modifications at various gene regulatory regions
In chapter 4, we describe why boolean models are good to learn histone modification networks. We describe the floating backward-forward feature selection algorithm which we used to select the histone modifications that showed the best dependency with the target modification. We then explain network selection algorithm, which was used to identify most prominent histone network from the selected histone modifications in the previous step.
Finally, in chapter 5, we summarize the work done in the thesis and discuss the future work that we will carry out.

Contributions of the thesis

The main contributions of the thesis are summarized below

1. A comprehensive list of dependencies of a histone modification with other modifications
2. Extensive profiles of histone modifications showing relative competition at various states of histone tails at TSS, enhancers, RNA polymerase II, insulators and predicted nucleosomes.
3. A list of new histone marks to characterize regulatory regions
4. A new computational approach to study histone code.

Chapter 1

Epigenetics and histone modifications

> *Geneticists study the gene; however, for epigeneticists, there is no obvious 'epigene'. Nevertheless, during the past year, more than 2,500 articles, numerous scientific meetings and a new journal were devoted to the subject of epigenetics. It encompasses some of the most exciting contemporary biology and is portrayed by the popular press as a revolutionary new science — an antidote to the idea that we are hard-wired by our genes. So what is epigenetics ?*
>
> — Adrian Bird
>
> *Gene++ ?*

Epigenetics is referred to as heritable modifications that can alter gene activity or expression without changes in DNA sequence. Epigenetic processes can effect the regulation of genes by changing the conformation of chromatin. These changes must last more than one cell cycle, for them to be considered as epigenetic processes.

The main epigenetic events are chromatin remodelling, histone modification and DNA methylation. Other interrelated factors that contribute to epigenetic events are histone variants, RNA modifications and RNA inference. The

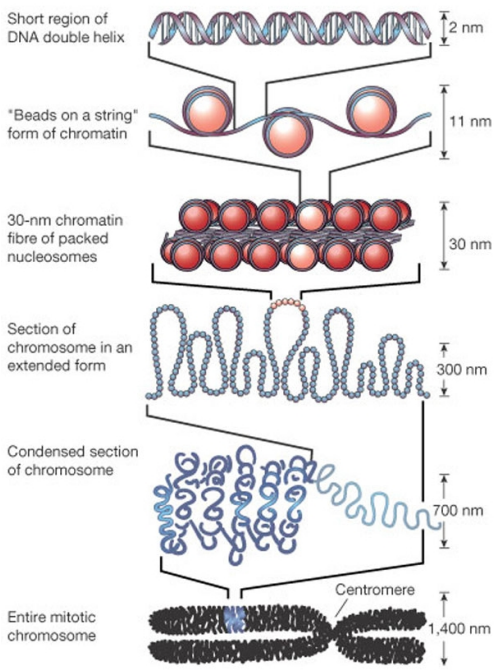

Figure 1.1: The different levels of chromatin organization [1]. The 11nm DNA wrapped around four histones represting beads on string model. DNA undergoes through further stages of compaction, finally resulting in 1.5μm condensed chromosome

focus of this thesis is on histones and post translational modifications that take place on their tails.

This chapter provides biological background required to understand histone modifications and is based on book *Epigenetics* by Allis, Jenuwein and Reinberg [2] . We start by providing a brief description of chromatin, followed by detailed description histone modifications and various post-translational modifications that take place on their tails.

1.1 Chromatin

DNA, within the chromosomes, is packaged into chromatin. Chromatin is basically DNA-nucleosome polymer, which consists of DNA, structural proteins and non-structural proteins. This DNA-nucleosome polymer is very dynamic in nature and can exist in many configurations. The repeating units of the

polymer, nucleosomes, are 147 base-pairs of DNA wrapped around an octamer of 4 histone protiens. These nucleosomes then form a higher order structures called chromatin.

In figure 1.1, we can see different levels of chromatin organization. Most of the time chromatin exists in 11nm fibre, often termed as "beads on a string". But as cell approaches mitosis or meiosis stage, DNA gets more tightly packed into 30nm solenoid and keeps on condensing till it takes the form of chromosome. The unwrapped DNA is approximately 2m in length, while in its tightest form its around 10μm (see fig 1.1). Now we look at the individual building blocks of chromatin.

The building blocks of chromatin

We can broadly divide chromatin into euchromatin and heterochromatin. Euchromatin is the the open and transcriptionally active part of the chromatin. The activating histone marks like histone acetylations, some histone variants are characteristic of euchromatin.

Heterochromatin is condensed, highly compact part of the chromatin. It also usually referred to as 'silent DNA'. The repressive histone marks and hypermethylated DNA are characterstic of heterochromatin. Heterochromatin can be further divided into constitutive and facultative heterochromatin. Constitutive heterochromatin are irreversibly silent, while facultative heterochromatin can become transcriptionally active again.

Now we briefly look at the building blocks of chromatin: the histone proteins, nucleosomes, linker histone and linker DNA.

The histone proteins

Histones are highly conserved, small, alkaline proteins that help to pack DNA and give it a shape. They are the main proteins of the chromatin and play a vital role in gene regulation. The five main classes of histones are H1/H5, H2A, H2B, H3 and H4.

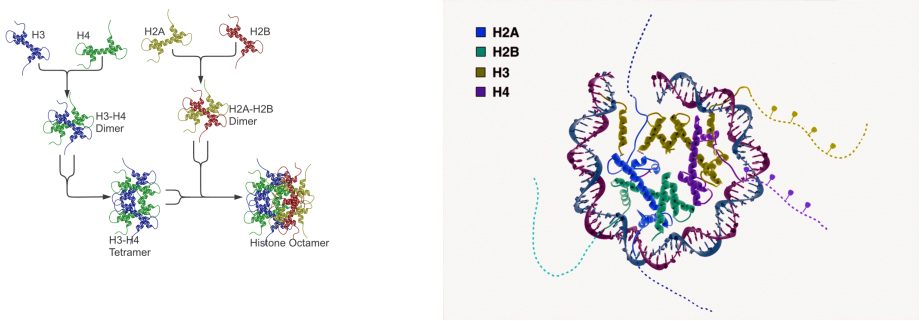

Figure 1.2: Schematic representation of formation of histone octamer from four core histone proteins and wrapping of DNA around this octamer [3, 4]

The core histone proteins

Histones H2A, H2B, H3 and H4 are known as core histone. Histones H2A/H2B and H3/H4 dimerise to form H2A-H2B and H3-H4 dimer respectively. After one more step of dimerization, H3-H4 dimer forms a tetramer, and then deposit on the DNA. This is followed by depositions of two separate sets of H2A-H2B dimer, resulting in formation of octamer of 4 core histone proteins (see figure 1.2). The four core histone proteins are very similar in structure and are highly conserved through evolution.

Nucleosomes

Nucleosome is fundamental repeating unit of chromatin. The nucleosome core is formed by two H2A-H2B dimers and tetramer of H3-H4. The term nucleosome refers to one nucleosome core particle and one of its adjacent linker DNA (see Figure 1.3).

The linker histone

Linker histone H1 is not part of a nucleosome but it helps in holding the nucleosome core and keeping the DNA wrapped around it. Apart from binding to nucleosome, the linker Histone H1 also binds to linker DNA. It has been speculated that linker histone determines the folding of chromatin [5]. Also the linker histone(H1) can exist in many isoforms. One of the largely found

isoform is H5.

The linker DNA

Linker DNA is double stranded DNA, around 8 to 114 bps, linking two nucleosomes together, in association with Histone H1. The linker DNA is the string in the "beads and string model". Linker DNA is not associated with open or transcribable DNA.

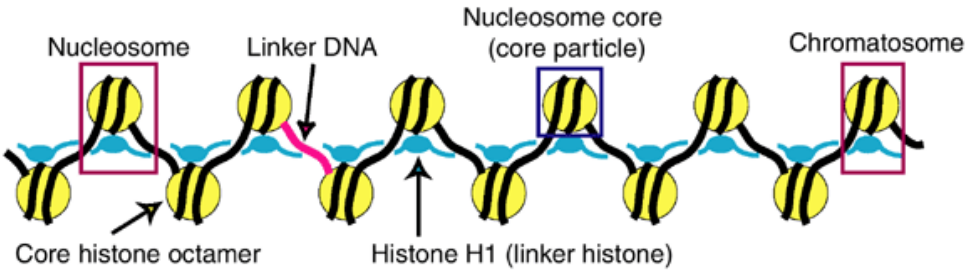

Figure 1.3: The chromatin representation: The core histone proteins are shown in yellow. The 147 bp of DNA(black) with the core histone proteins forms core nucleosome. The core nucleosome with linker histone H1(blue) forms chromatosome. Two nucleosomes are linked with linker DNA(red) [6]

1.2 Post-translational modifications on histones

In this section, we look at various post-translational modifications that take place on the tails of histone proteins. We start by explaining what charges DNA and histones carry and why DNA wraps around the histone proteins. After this we move on the tails of histones and explain which amino acids are susceptible for post-translational modifications and various types of modifications that can take place on them.

The charges on DNA and histone proteins

DNA is negatively charged while histones being highly basic proteins carry a positive charge.

DNA is made up of a nucleobases (adenine, thymine, guanine, cytosine), a five carbon sugar (either ribose or 2-deoxyribose), and a mono-, di-, or tri-phosphate group. Now the phosphate group is readily soluble in water, and so DNA acquires a negative charge by dissociating into phosphate esters(see equations below)

$$H_3PO_4(l) + 2H_2O(l) \Rightarrow HPO_4^{-2}(aq) + 2H_3O^+(aq) \tag{1.1}$$

$$RH_2PO_4(l) + 2H_2O(l) \Rightarrow RPO_4^{-2}(aq) + 2H_3O^+(aq) \tag{1.2}$$

Hence we can speculate that negatively charged DNA will readily wrap around anything that is positive to stabilize itself. So it warps around the positively charged histone proteins, leading to formation of nucleosomes.

Histone tails

Amino acids susceptible for post-translation modifications

Histone tails are susceptible to many post-translational modifications, but not all the amino acids sites on the tails are susceptible for post-translational modifications. Lysine(K), Arginine(R), Serine(S) and Tyrosine(Y) are most susceptible for PTMs.

Lysine(K) and Arginine(R) are amino acids with electrically charged side chains and usually carry a positively charged N terminal. Serine(S) is an amino acid with polar uncharged side chains, while Tyrosine(Y) is a amino acid with hydrophobic side chain. In Figure 1.4, we can see various post-translational modifications that take place on four core Histone proteins H2A, H2B, H3, H4.

We can broadly divide the post-translational modifications into 2 groups

Small chemical group PTMs

- Acetylation
- Methylation
- Phosphorylation

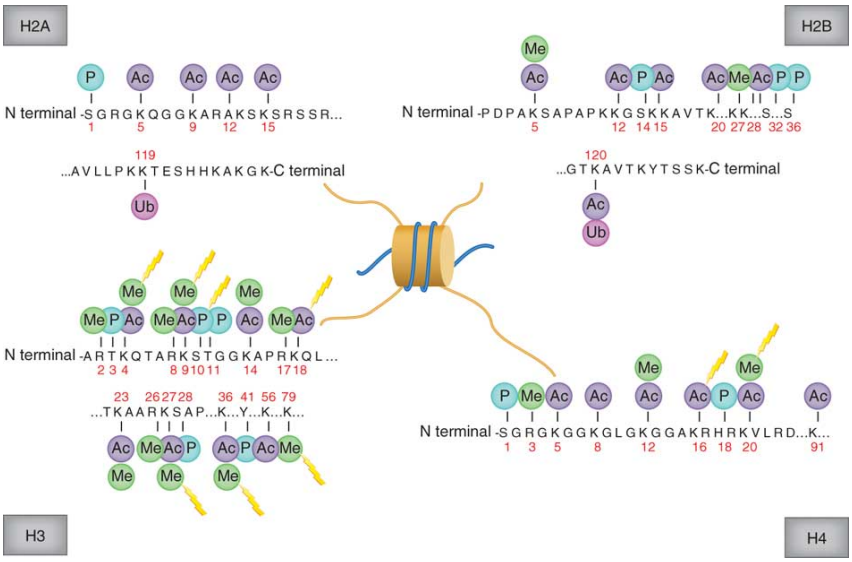

Figure 1.4: Various histone tails protruding from histone octamer. The following amino acids: Lysine(K), Arginine(R), Serine(S) and Tyrosine(Y) are most susceptible for post translational modifications [7].

Large peptide PTMs

- Ubiquitylation
- Sumoylation

Various post-translational modifications that take place on histone tails

Acetylation

Acetylation exclusively happens at lysines and it was the first discovered, and the most studied histone modification. Acetylations are usually associated with active gene transcription, but some acetylated lysine residues have shown association with marks known for repressed gene transcription.

Enzymes which transfer acetyl marks at lysine sites are called HATs (Histone Acetyl transferases). Enzymes which remove acetyl marks are called HDACs (Histone DeACetylases). Both HATs and HDACs are recruited by DNA-binding transcription factors. The activators recruit HATs, which acety-

late the histone tails and supposedly loosens the chromatin, while repressors recruit HDACs which deacetylate the tails and again making the lysines positively charged, hence tightening the chromatin(see figure 1.5). A more detailed explain can be found at [8].

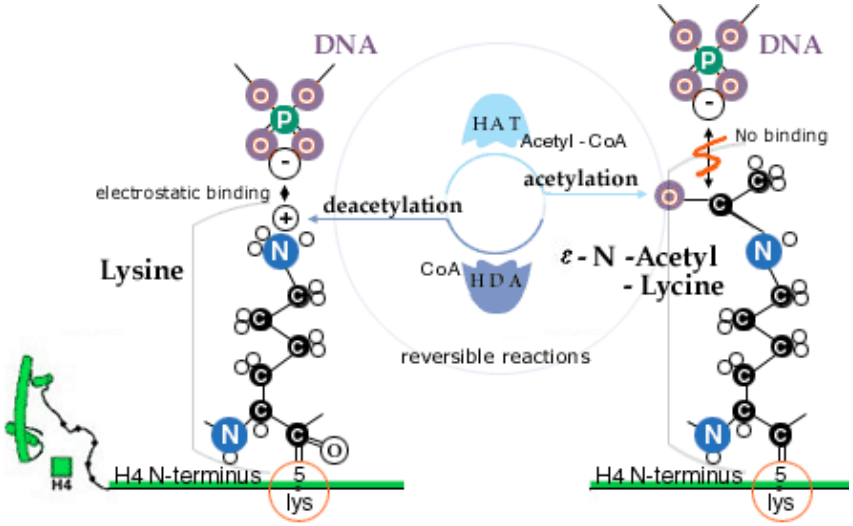

Figure 1.5: Acetylation and deacetylation of lysines by HATs and HDACs. When lysines are acetylated, the acetyl group blocks/prevents the electrostatic interaction between DNA and histone tails, hence opening the chromatin. Thus active genes are associated with hyperacetylated chromatin, while inactive genes are found at hypoacetylated chromatin [8].

The whole process can be summarized as: Lysine which is positively charged, readily binds with negatively charged DNA, but now due to acetyl group, lysine does not have a positive charge and hence, its interaction with negatively charged DNA is affected, resulting in loosing of the chromatin.

Methylation

Methylation is the most complex of all histone modifications, and it can act as both active and repressive mark. Methylation occurs at varying degrees on lysine and arginine. Lysine can be mono-, di-, tri- methylated while arginine can be mono- and di- methylated(see figure 1.6).

Methylation is a stable modification, and is conserved across many cell generations. There are enzymes which can attach/remove methyl groups from

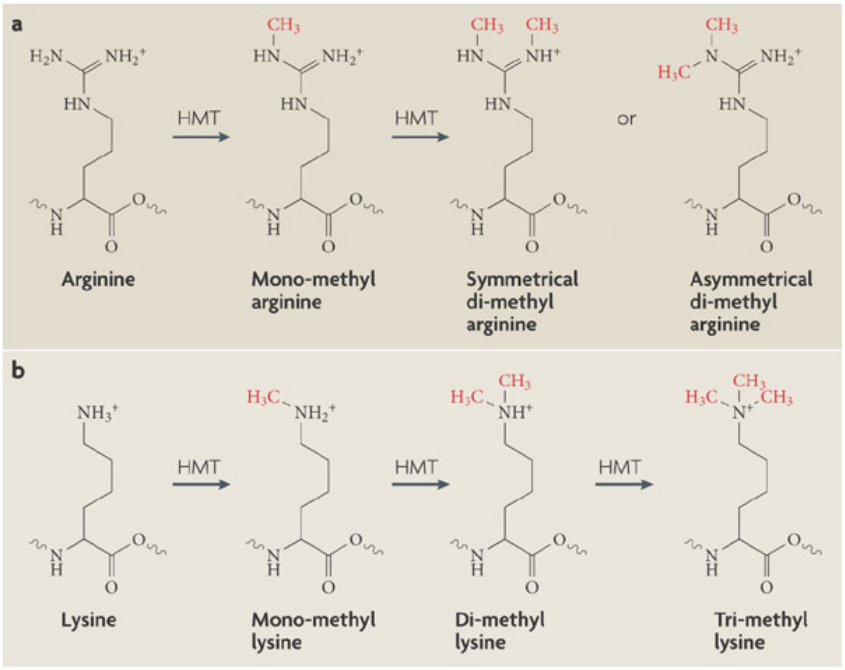

Figure 1.6: Varying degree of methylations on lysine and arginine residues. Lysine can be mono-, di-, tri- methylated while arginine can only be mono- and di- methylated [9]

lysine. Histone methyl transferases (HMTs) transfer methyl groups to lysine and arginine sites while Histone DeMethyltranferases (HDMTs) remove the methyl groups from lysine and arginine sites.

We study various lysine and arginine methylations at different regulatory regions in chapter 3.

Phosphorylation

The most significant phosphorylation is on histone 3 serine 10 (H3S10ph). The exact mechanism of phosphorylation is still unknown but the following model is most accepted. First phosphorylation introduces a negative charge and the DNA is also negatively charged, so it reduces the affinity for DNA. H3S10ph also causes ejection of HP1. HP1 is heterochromatin protein, whose function is gene repression by heterochromatin formation. Hence ejection of HP1 causes promoter activation [2] .

Ubiquitylation

Ubiquitylation and Sumoylation are large peptide modifications. They are a bit similar in structure but different in charges. Ubiquitylation can be both active and repressive modification.

MonoUbiquitylation on Histone 2B lysine 123 is highly conserved and leads to methylation of Histone 3 lysine 4 and is associated with transcriptional activation. While MonoUbiquitylation on Histone 2A lysine 119 is associated with transcriptional repression [2].

Sumoylation

Sumoylation is again a highly conserved and is connected with repression of gene transcription. Sumoylation blocks lysine substrates and hence, methylases and acetylases cannot reach lysine and modify it. Also it is believed that sumoylated lysines recruit HDACs and lead towards repression of gene transcription [2].

The histone variants

The replacement of canonical S-phase nucleosomal histones by variants can lead to differentiation of chromatin, independent of replication. H2A.Z which is one of the histone variants that is used in this thesis, is associated with transcriptional regulation. H2A.Z is not uniformly spread throughout the genome and prevents spreading of silent chromatin [2].

1.3 The Histone cross talk

Attempts have been made by researchers to come up with a code for histone modification like one similar to DNA code of 4 bases to 20 amino acids. But the communication between histone modifications is not as straight forward as DNA language. Most of the chromatin modifications are very dynamic in nature and highly influence each other.

In Figure 1.7, we can see cross talk among Histone tails. As an example of trans cross talk, we can see that ubiquitylation of histone 2B lysine 123

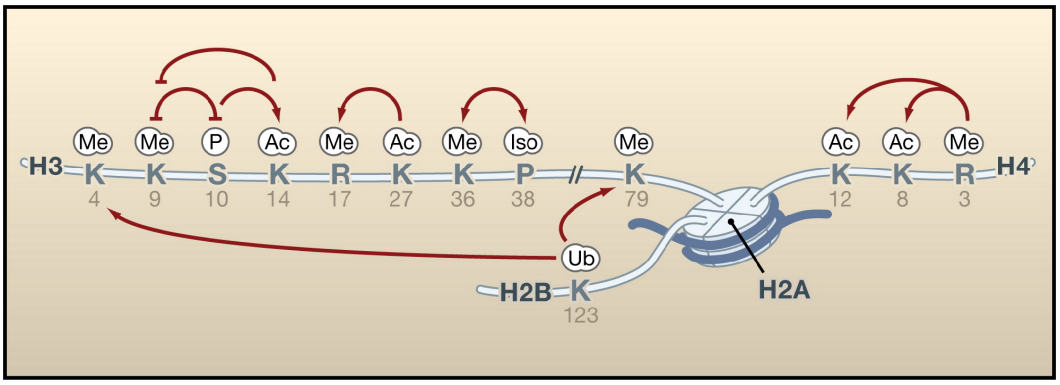

Figure 1.7: Different cis and trans patterns of Histone cross talk [10]

promotes methylation of histone 3 lysine 4 and histone 3 lysine 79. As an example of cis cross talk, we can see that phosphorylation of histone 3 Serine 10 promotes acetylation of Histone 3 lysine 14, which represses methylation of histone 3 lysine 9, which in turn represses its own phosphorylation [2]. So we can see that cross talk on the tails of histone proteins can be very complex.

Cross talk of histone modifications is considered as Histone code and this thesis makes a attempt to decode it.

Chapter 2

Histone modification data processing

> *It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts.*
>
> — Sherlock Holmes
>
> *Is H3K4me3 an activating mark ?*

In this chapter, we explain the steps that we followed to process histone modification data. We start by explaining how the reads are sequenced using ChIP-seq protocol. Next, we briefly explain the software's that we used to map the short reads and identify nucleosome-positions. After this we explain how we assigned the histone modifcations to the identified nucleosomes, and also show correlation and clustering results.

2.1 ChIP-Sequencing

ChIP-seq stands for Chromatin Immuno-Precipitation plus Sequencing and is one of the common methods for measuring epigenomic signals. It is an experimental design to study protein-DNA interactions, in order to identify enriched genomic sites where proteins bind to the DNA. These proteins can be transcription factors, histones, or the tails of histones where modifications take place. This method when combined with high-throughput sequencing is

termed as ChIP-seq. The procedure of ChIP-Seq is as follows(see figure 2.1)

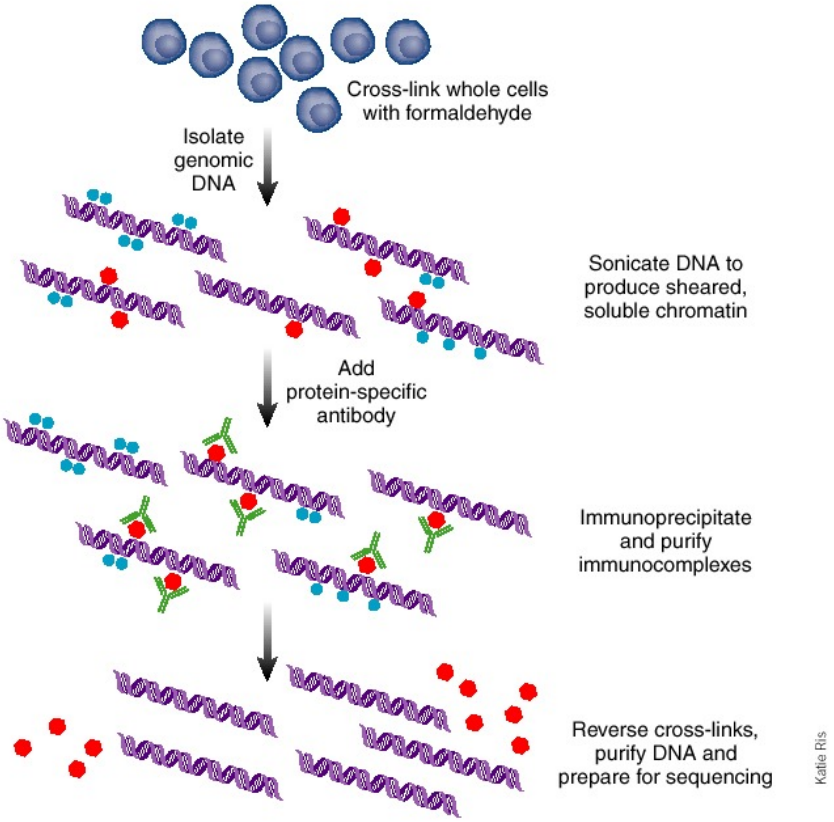

Figure 2.1: Various stages of chromatin immunoprecipitation [11]

Cross-linking: The proteins are cross linked with DNA using formaldehyde. Formaldehyde is a popular cross-linking agent which helps in forming covalent bonds between DNA and proteins.

Sonication - Now the DNA is isolated from cell nucleus and randomly cut into small fragments by sonication or enzymes.

Immunoprecipitation: An antibody specific to the epigenetic mark of interest is added to pull out the isolated protein and the attached DNA fragments.

Reverse crosslinking: The crosslinking between the protein and DNA is reversed and the DNA fragments specific to epigenetic marks are purified.

Sequencing: To identify these DNA fragments one can do massive parallel high-throughput sequencing(Seq part of ChIP-Seq), during which sequence tags 25-75 bps long, from the end or beginning of a fragment are identified.

The total number of reads depend upon sequencing depth, and are usually on order of millions for one ChIP-seq experiment.

Now the short reads are aligned back to the reference genome, which we will discuss in the following section.

2.2 Mapping short reads

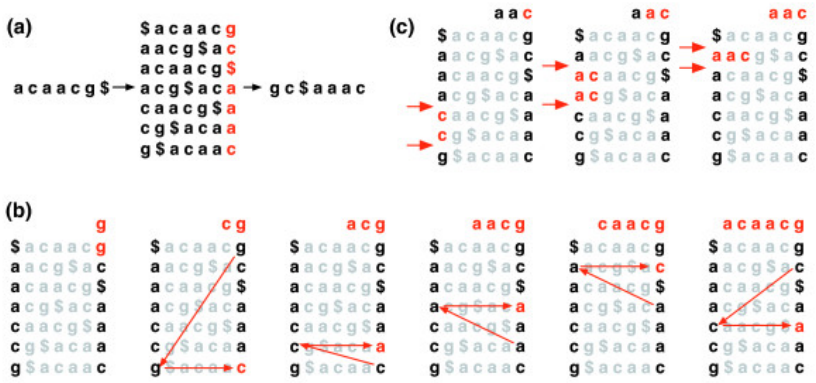

Figure 2.2: Burrows-Wheeler transform: The approach bowtie uses to map short reads. **(a)** For sequence 'acaac', the Burrows-Wheeler matrix and transformation. **(b)** Last first (LF) repeatedly applied to get the original sequence. **(c)** How to identify a set of suffixes prefixed by 'aac' [12].

The two popular approaches to match short reads to a genome are hashtable-based approach(for example MAQ) and Burrows-Wheeler transformation(for example Bowtie). We used Bowtie to map histone modifications short read. We now briefly discuss the details of Burrows-Wheeler transformation, which is illustrated in figure 2.2.

The main advantages of BWT is to produce blocks of strings with long runs, in order to increase the compressibility of the data, but this increase is at the expense of time and system resources. Basically, BWT consists of a transformation and re-transformation step.

We first add a special character($ in this present case) to the string and generate all the cyclic shifts of the string. Now we sort the string lexicographically i.e. in alphabetical order, with the special character getting the highest

preference. The last character of each row contains the transformed genome. Figure 2.2a shows an example of such an transformation when applied to string "acaacg". Figure 2.2b shows how to recover the original text by last-first(LF) mapping. The i^{th} occurrence of a character in the first column corresponds to the i^{th} occurence in the last column. Figure 2.2c shows how to identify the set of reference suffixes. The key properties of Burrows-Wheeler transformation are that all occurrences of same suffix are next to each other, and are not scattered throughout the genome (see figure 2.2c) .

While mapping the reads we set the seed length equal to read length which was 36 bps in order to get better alignment. Also we only allowed 2 mismatches in the seed. The alignment of all histone modification was done against human genome 18, and only uniquely mapped reads were considered. We used the following command

```
./bowtie -t -l 36 -m 1 /hg18 input.file output.map
```

A description of parameters used during mapping

-t	timing statistics. Print the amount of wall-clock time taken by each phase
-m 1	how many legal alignments should be reported
hg18	the data was aligned to this version of the human genome
-n 2	number of mismatches allowed in the first L bases. The first L bases are called the "seed"
-l 36	the seed length, we set it equal to the size of the reads to get better alignments

The datasets, we used in this thesis were from studies [13, 14]. The histone acetylations and methylations short reads can be downloaded from `http://www.ncbi.nlm.nih.gov/sra` with accession numbers SRA000206 and SRA000287 respectively. This data can also be downloaded from
`http://dir.nhlbi.nih.gov/papers/lmi/epigenomes/hgtcell.aspx`
`http://dir.nhlbi.nih.gov/papers/lmi/epigenomes/hgtcellacetylation.aspx`

2.3 Identifying the nucleosome positions

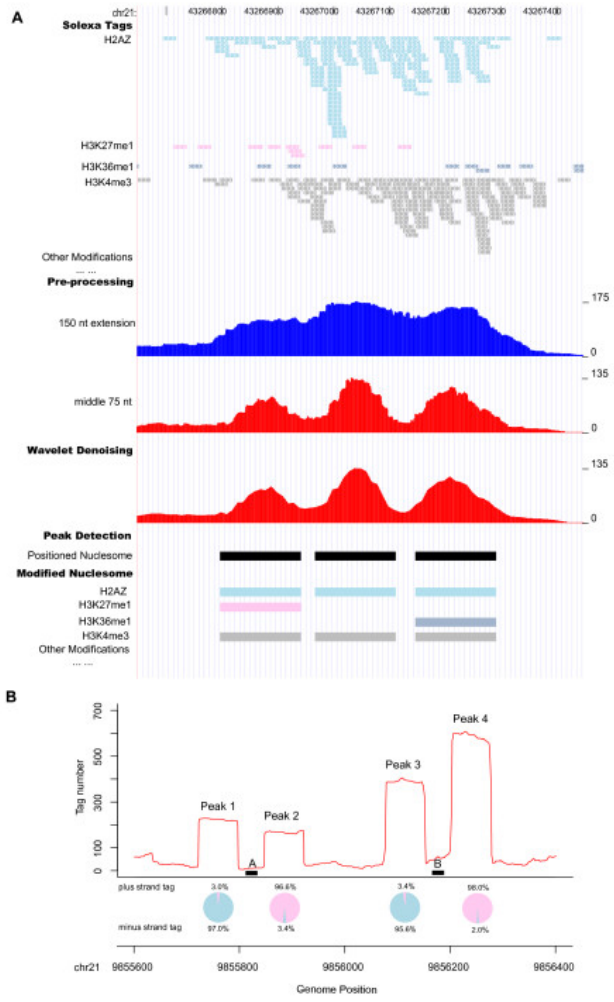

Figure 2.3: Identification of nucleosomes from ChIP-Seq data. **(a)** Each read is extended to 150 nt (blue peaks) and then middle 75 nt (red peaks) are taken. These peaks are then smoothed in order to get better signal resolution. **(b)** Most detected peaks had equal number of tags on positive and negative strand, but some peaks had unbalanced tag counts. Such peaks were discarded from further analysis in NPS [15]

We used method developed by [15] to identify nucleosomes from the combined histone modification data. The method uses a combination of wavelet

denoising to smooth the signal and Laplacian of Gausssian(LoG) technique to detect the nucleosome positions. Nucleosomes follow a periodic pattern. Usually 6-7 nucleosomes occur together forming a nucleosomal array with +1 nucleosome most enriched. This periodic behaviour represents a continuous wave-form of enriched histone modifications/nucleosomes. The edges of these enriched peaks are then detected using Laplacian of Gaussian (LOG) edge detection technique.

In this method each tag is extended to 150 nt and then middle 75 nt are taken in order to get better signal resolution. We can see in Figure 2.3a the difference in resolution of peaks, when only extension is considered(blue) and when middle 75nt are taken(red). The smoother peaks below are result of wavelet denosing. Most detected peaks had equal number of tags on positive and negative strand, but some peaks had unbalanced tag counts, where number of tags on one strand were four times higher than on other strand(Figure 2.3b). Such peaks are discarded from further analysis in NPS. Next peaks whose width was less than 80 nt or greater than 250 nt were removed. Using the software, we identified around 750,000 nucleosomes and then selected top 186,492 nucleosomes for further analysis.

2.4 The binary histone modification data

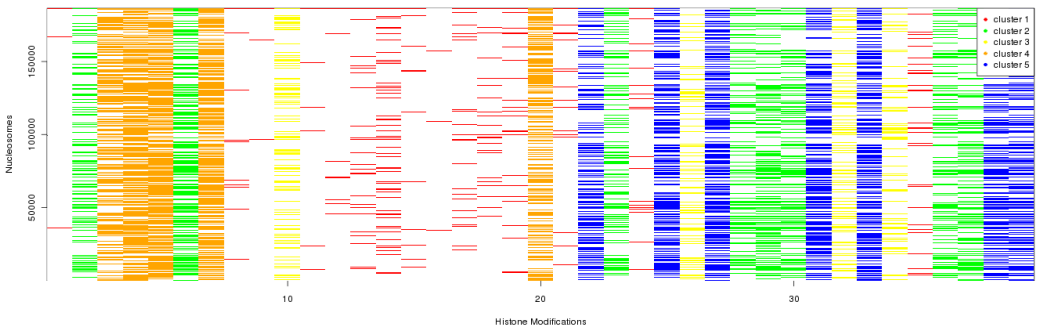

Figure 2.4: Assignment of histone modifications to Nucleosomes: The x-axis shows 39 histone modifications and the y-axis show the predicted nucleosome. If a modification is present inside a nucleosome, then it is represented by a color bar else a blank.

After identifying nucleosome positions, we assigned histone modifications to the nucleosomes. We counted total number of tags for a particular modification in the nucleosomal peak region and then compared it with expected number of tags for that chromosome using Poisson distribution. The Poisson distribution with parameter λ is described as

$$P\{N = j\} = \frac{\lambda^j}{j!} e^{-\lambda} \tag{2.1}$$

We assigned a histone modification to a nucleosome if the read count p-value was less than 10^{-3} in the nucleosome peak region with respect to the chromosome to which that nucleosome belonged . Finally we ended up with binary matrix of 0's and 1's. In Figure 2.4 the y-axis represents the nucleosomes and the x-axis represents the 39 histone modifications. If a modification is assigned to a nucleosome, then it is represented by a colour bar, else it is blank. The color represent the five clusters as found using k-means algorithm.

2.5 The correlation plot of histone modifications

In figure 2.5, we can see the correlations among various histone modifications. A 45° blue line represents maximum positive correlation i.e. +1 and a 135° represents maximum negative correlation i.e. -1. All intermediate correlations are represented by colours that vary from blue to red. A perfect white circle represents zero correlation.

In this figure 2.5, we can clearly two clusters. First at the top-left corner, where H3K4me3 marks is seen to have positive correlation with a lot acetylation marks. Second, we can see at bottom-right corner of the figure 2.5, where a lot of repressive marks have clustered together showing positive correlation. We used **plotcorr** R function to plot the correlation matrix of histone modification data.

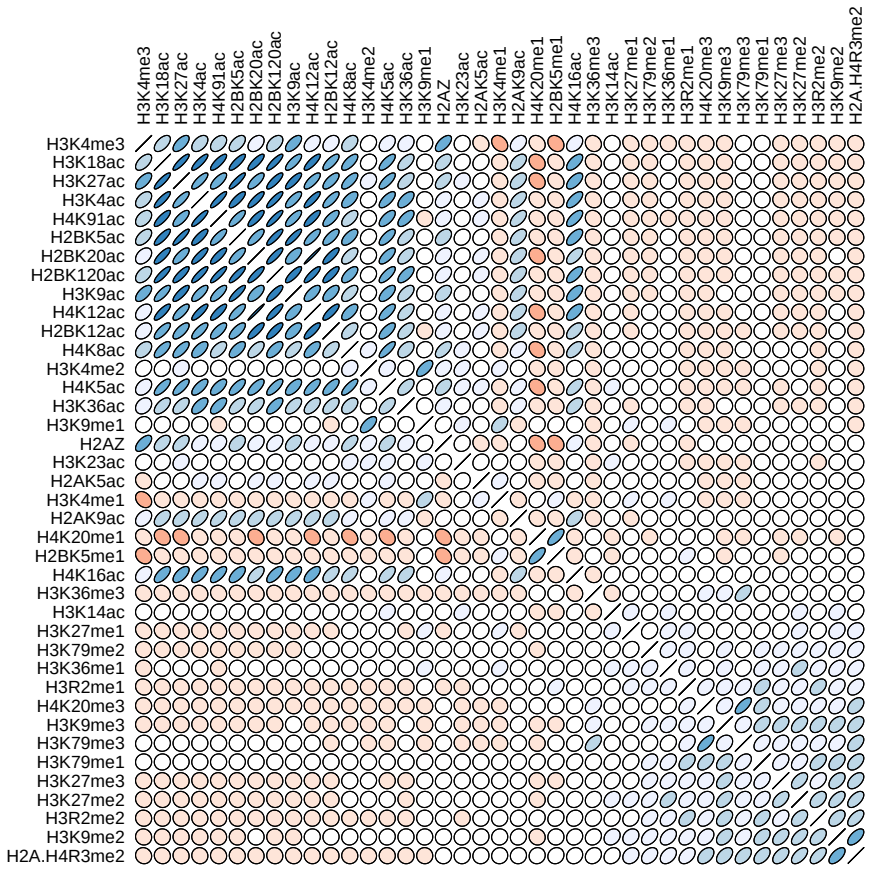

Figure 2.5: The correlation plot of histone modifications: The positively inclined blue represents positive correlation and negatively inclined red represents negative correlation. A perfect white circle represents zero correlation. We can clearly see two clusters one at top left corner where a lot of acetylation marks show positive correlation with H3K4me3 and H3K27ac mark, and one at bottom right corner where some methylation marks show positive correlation with H3K27me3 and H3K79me3

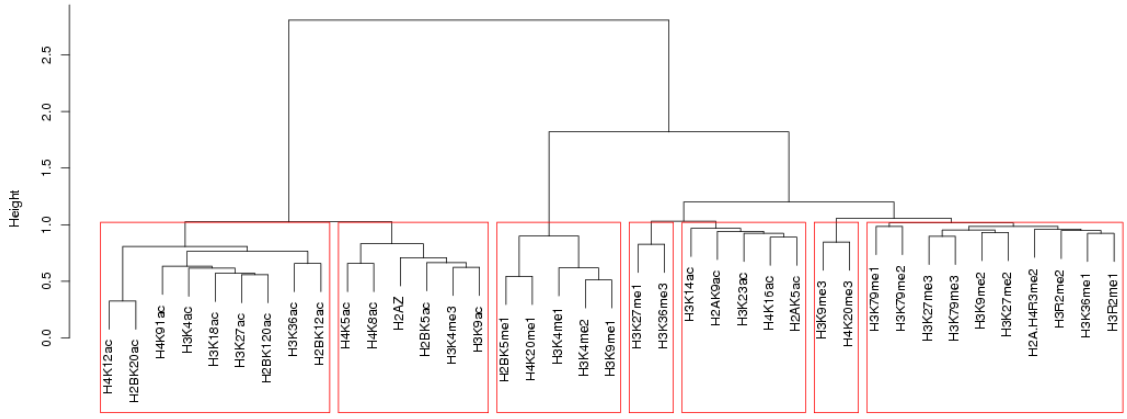

Figure 2.6: The hierarchical clustering of histone modifications. The clustering result complements well with the correlation plot result where a lot of acetylation marks have clustered together with H3K4me3 mark, while a lot of methylations have clustered together with repressive marks such as H3K27me3 and H3K9me3

2.6 Hierarchical clustering of histone modifications

In Figure 2.6, we can see that all repressive marks have clustered together in the last cluster. We can also H3K36me3 showing a close association with H3K27me1 mark. In another cluster 3, we can see that H3K4me1 mark, a known enhancer mark, has been clustered with H3K4me2, another enhancer mark, but two most interesting marks, which can also be classified as enhancer marks, in this cluster are H4K20me1 and H3K9me1. Also we can see that a lot acetylation marks have clustered together, leading to the idea of kinetic proof reading for chromatin condensation and de-condensation.

The hierarchical clustering was based upon Pearson correlation. First a distance matrix was calculated for histone modification data. Initially each object was assigned to its own cluster, and then **hclust** function by F. Murtagh was used, which after each iteration joined two most similar clusters.

Chapter 3

Gene regulation and profiles of histone modifications at various regulatory regions

> *Every person is important. It doesn't matter whether you play the violin, the flute, the cello, or the drums, you're still part of the orchestra*
>
> — ANONYMOUS
>
> *Who is the conductor here ?*

An orchestrated set-up of regulatory elements and transcriptional machinery is required for controlled and successful expression of a gene. A disharmony among musicians of an orchestra can lead to unpleasant music, similarly a small defect or mutation in any of the components of transcriptional set-up of a gene can lead to cascade of effects that can result in uncontrolled or wrong expression of a gene.

In this chapter, we discuss and visualize 39 chromatin modifications at the various gene regulatory regions such as promoters, enhances, insulators, and at predicted nucleosomes and TSS. We start by briefly looking into various gene regulatory regions and the associated histone modifications.

3.1 Transcriptional machinery and histone modifications

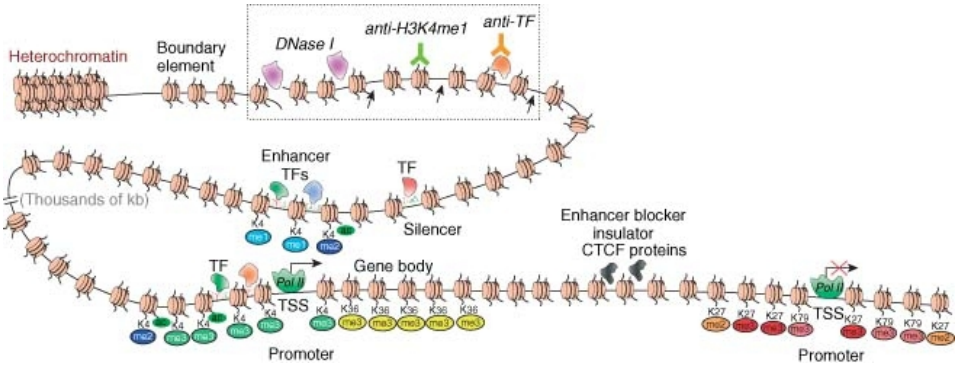

Figure 3.1: Histone modifications, characteristic enzymes and transcription factors at various gene regulatory regions [16]

Not long ago, a lot of scientist believed that chromatin was not important in gene regulation, but recent progress in chromatin biology has shown that chromatin signatures can define various regulatory regions, and is considered as one of the major emerging concepts to study gene regulation. Nucleosome-positioning at most regulatory regions like promoters, enhancers, insulators can be well characterized by presence of various histone modifications and histone modifications in turn can be used to define various regulatory regions, for example, in [17, 18] the enhancer locations were determined using histone modifications and then these profiles were trained to identify new enhancers. In [19], transcription factor binding locations were identified by flanking nucleosomes i.e. nucleosome pairs with center position 250-450 bps apart. We now have characteristic histone marks for most of the regulatory regions, and can annotate most of the genome with them, like we do with characteristic enzymes and transcription factors. For example [20, 21]:

Promoters	RNAPII, TFIID, H3K4me3
3' end of coding region	H3K36me3
5' end of coding region	H3K9ac, H3K14ac, H4K12ac
Gene repression	H3K4me0(unmethylated H3K4)
Enhancers	STAT1, P300, H3K4me1, H3K4me2, H3K27ac

Insulators	CTCF, H3K27me3

3.2 Histone modifications at predicted nucleosomes

Nucleosomes are 147 base-pairs of DNA wrapped around the proteins. The proteins in this case are not transcription factors but an octamer of 4 histone protiens H2A, H2B, H3 and H4. These nucleosomes then form a higher order structures called chromatin. We can view nucleosomes as polymers of small repeating units which consists of approximately 1.7 turns of DNA and eight histone proteins.

Apart from compacting extremely long DNA molecule, nucleosomes play a extremely vital role in accessibility of DNA and in-turn regulation of genes. Nucleosomes in themselves are not very important but a missing or a displaced nucleosome is a indication of presence of a transcriptional element. For example, throughout the genome it has been seen that promoters and other regulatory sequences are mostly nucleosome depleted while most of the transcribed regions are characterized by well positioned nucleosomal arrays.

Recent evidences [22, 23] have claimed nucleosome-positioning and enhancer-DNA looping or long range DNA interactions as two major concepts to study gene regulation, which makes a thorough dissection of nucleosomes at various regulatory regions even more important. In this section, we show profiles of histone modifications at the predicted nculeosomes. We used **R** programming language and **ShortRead** and related bioconductor [24, 25] packages to generate these plots.

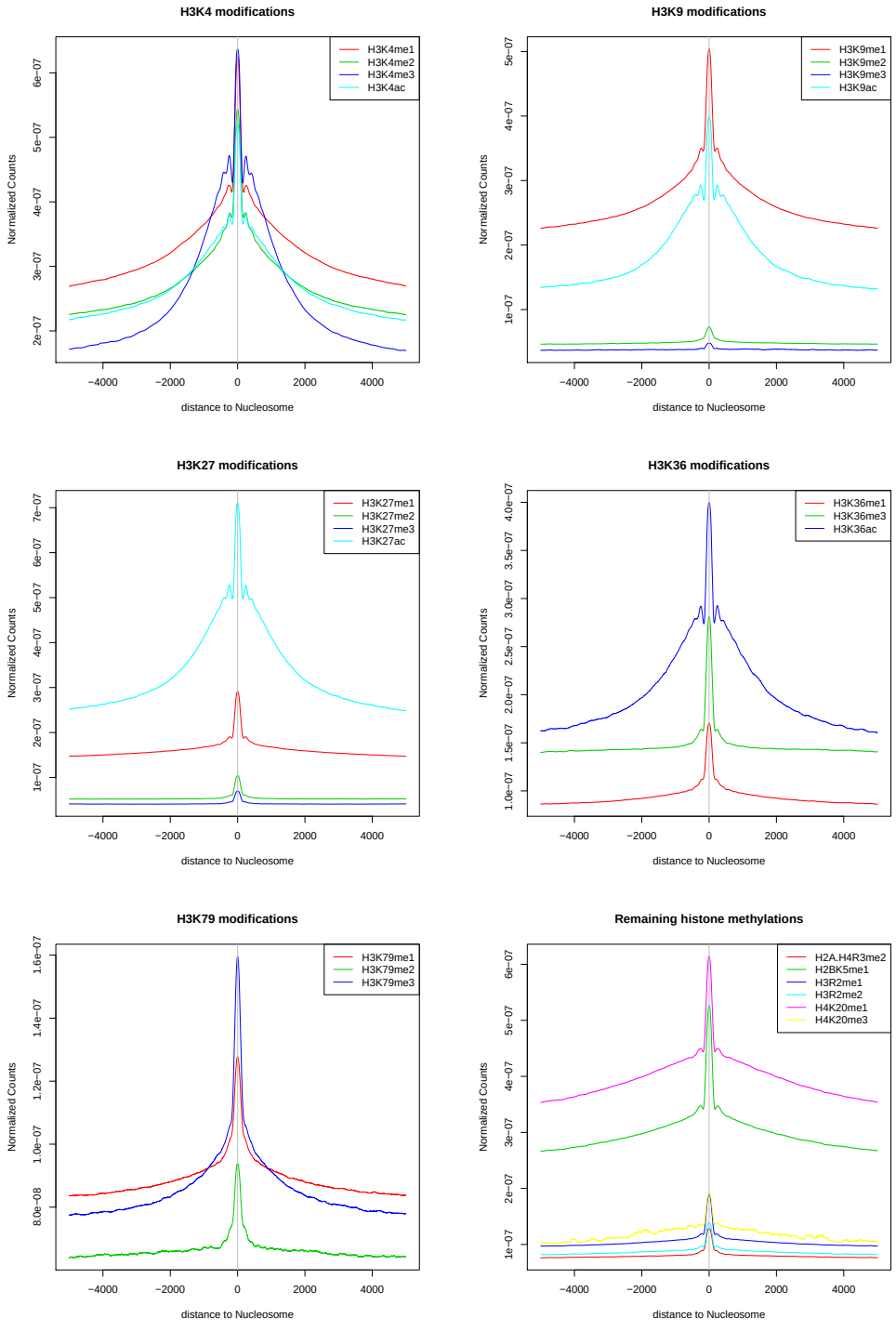

Figure 3.2: Histone modifications profiles at predicted nucleosomes. The nucleosomes positions were predicted by combining 39 histone modification data. We can see show how well histone modifications at a specific site compete with one another.

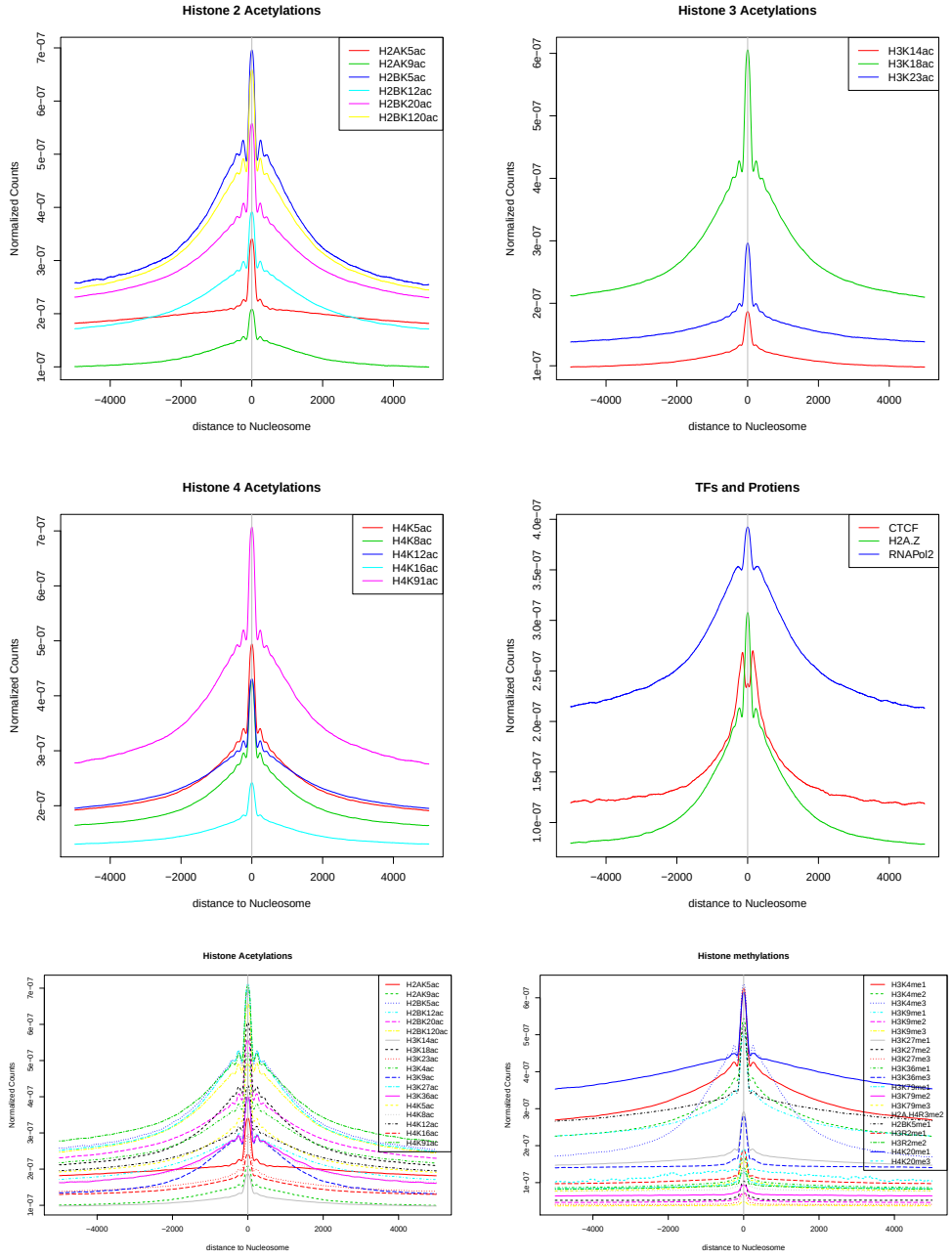

Figure 3.3: Of the available acetylations; H3K27ac, H2BK120ac and H2BK5ac were most enriched. Of the available methylations; H4K20me1, H3K4me3, H3K4me2 and H3K4me1 contributed most to the nucleosomes

H3K4 modifications: All four states of H3K4 were elevated at the predicted nucleosome positions. H3K4me3 had distinct ±1, ±2 peaks w.r.t to central peak, while H3K4me1 and H3K4me2 didn't showed ±2 peaks, but ±2 peaks were slightly elevated for H3K4ac.

H3K9 modifications: H3K9me2 and H3K9me3 showed almost no elevation. This might be due to the fact that didn't had enough sequenced reads. H3K9me1 and H3K9ac showed distinct elevation for ±1 peaks, with H3K9ac also showing slight elevation for ±2 nucleosomes.

H3K27 modifications: H3K27ac only showed high elevation at predicted nucleosomes with ±1 and ±2 nucleosome peaks. H3K27me1 showed slight elevation at the predicted positions.

H3K36 modifications: H3K36ac showed maximum elevation at nucleosomes and had distinct ±1 nucleosome peaks, followed by H3K36me3 and H3K36me1.

H3K79 modifications H3K79me3 showed maximum elevation followed by H3K79me1 and H3K79me2. None of them showed distinct ±1 peaks.

Remaining Histone methylations: H3K20me1 and H2BK5me1 were two mono-methylation which showed any enrichment at nucleosomes , in addition to H3K4me1. Others showed little enrichment at the nucleosomes and lacked ±1 peaks.

H2 Acetylations: All the acetylations at Histone 2 were enriched at nucleosomes to varying degrees, with distinct ±1 and ±2 peaks. H2BK120ac and H2BK5ac showed most enrichment.

H3 Acetylations: H3K18ac was most interesting of remaining H3 acetylations, with distinct ±1 and ±2, and showing some tendencies for ±3. Previous studies [14] have shown its relation to H3K27ac, and it had profile quiet similar to H3K27ac. Its relation is also obvious from high correlation in the correlation plot(see figure 2.5).

H4 Acetylations: All H4 acetylations had similar patterns except H4K16ac which showed little enrichment for ±2 nucleosome peaks

TFs and Proteins: CTCF had a dip at nucleosome positions but showed showed some elevation either side of the nucleosome. H2A.Z had similar profile as other histone modifications. RNApol2 was more elevated as compared to CTCF and H2A.Z

3.3 Histone modifications at TSS

Transcription start site(TSS) is the site where transcription of gene into RNA begins. In eukaryotes, the transcription of gene is done by RNA polymerase II(RNAPol2). We look various histone modification at RNAPol2 binding sites in the next section. Factors involved in transcription of genes by RNA polymerase II can be classified into 3 groups: general transcription factors(GTFs), activators and co-activators. GTFs assemble on core promoter to assemble pre-initiation complex(PIC), which directs RNA polymerase II to the transcription start site [26].

We got the list of transcription start sites from `http://hgdownload.cse.ucsc.edu/goldenPath/hg18/database/refGene.txt.gz` and then plotted all the histone modifications at these positions. It has been shown in previous studies that some histone marks like H3K4me3 and H3K27ac have specific patterns and profiles around TSS. Hence if specific combination of histone marks can be found that can be characteristic of TSS, then it would help in identifying novel Transcription start binding sites(TSS). We now show profiles of histone modifications at transcription start sites.

H3K4 modifications: H3K4me3 showed the characterstic and the expected +1 nucleosome more elevated than -1 nucleosomes, with a sharp dip at TSS. H3K4ac showed similar patterns as H3K4me3 but was less elevated. H3K4me2 and H3K4me1 showed rise in peaks but were much further as compared to +1 peak of H3K4me3, giving indication of some regulatory elements like enhancers.

H3K9 modifications: H3K9ac showed a similar profiles as H3K4me3 but was less elevated, while profile of H3K9me1 was similar to that of H3K4me2. H3K9me2 and H3K9me3 showed no enrichment at TSS.

H3K27 modifications: H3K27ac was most elevated either side of TSS. H3K27me1 had sharp dip at TSS but showed modest elevation downstream of TSS. H3K27me2 and H3K27me3 showed no elevation at TSS.

H3K36 modifications: H3K36ac was elevated either side TSS with a higher -1 nucleosome peak than +1 nucleosome peak. H3K36me1 was a bit similar in nature to H3K27me1, while H3K36me3 showed elevation downstream of TSS, which a indication that H3K36me3 mark is enriched at gene regions.

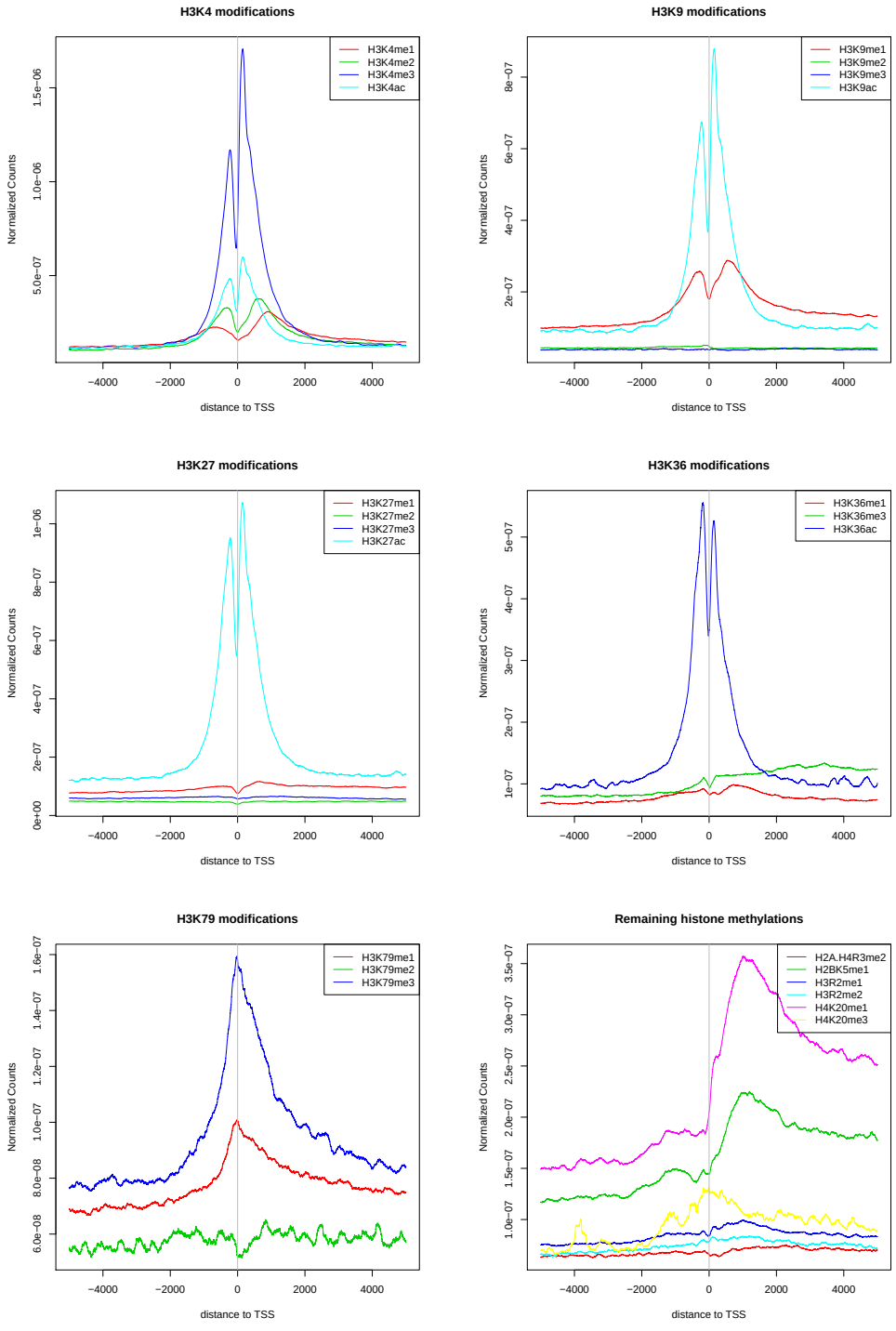

Figure 3.4: Histone modifications profiles at TSS

H3K79 modifications Of the modifications at H3K79 state, the profiles of H3K79me3 and H3K79me1 were same but H3K79me3 was more elvated. H3K79me2 showed some peaks but was not significantly elevated.

Remaining Histone methylations: H4K20me1 and H2A.H4R3me2 had similar profiles at TSS but H4K20me1 was more elevated. H4K20me3 showed most strange profile, with some enrichments both upstream and downstream of TSS.

H2 Acetylations: H2BK5ac and H2BK120ac had similar profiles and were most elevated around TSS, while H2AK9ac, H2BK20ac and H2BK12ac also had similar profiles but were less elevated. H2AK5ac showed no ± 1 peaks around TSS instead had a sharp dip at TSS.

H3 Acetylations: H3K18ac was most elevated around TSS and had a profile similar to H3K27ac. H3K14ac and H3K23ac had similar profiles but had very less elevation.

H4 Acetylations: H4K91ac was most elevated around TSS and had a profile similar to H3K27ac. H4K5ac, H4K8ac and H4K12ac had all same profiles around TSS. H4K16ac also showed some enriched around TSS

TFs and Proteins: RNAPol2 as expected showed a high enrichment a few base-pairs ahead of TSS. H2A.Z showed ± 1 nucleosomal peaks around TSS. CTCF also showed some enrichment at TSS.

3.4 Histone modifications at RNA Pol II

RNA polymerase II elongation complex is formed after a series of steps such as promoter melting, clearance and escape. After RNA polymerase escapes from the promoter, various auxiliary components like TFIID, TFIIE, TFIIH and mediator remain at core promoter. Now the transcription of genes only requires RNA polymerase II-TFIIF and TFIIB. Thus, the promoters regions can be mapped by localization of regulatory elements such as RNA polymerase II(RNAPol2) and transcription initiation factor TFIID-associated factor 1(TAF1), and also by histone marks such as Histone 3 lysine 4 trimethylation(H3K4me3) [21]. In this section, we show histone modification profiles at RNAPol2 binding sites. We got the RNA polymerase binding site from [13].

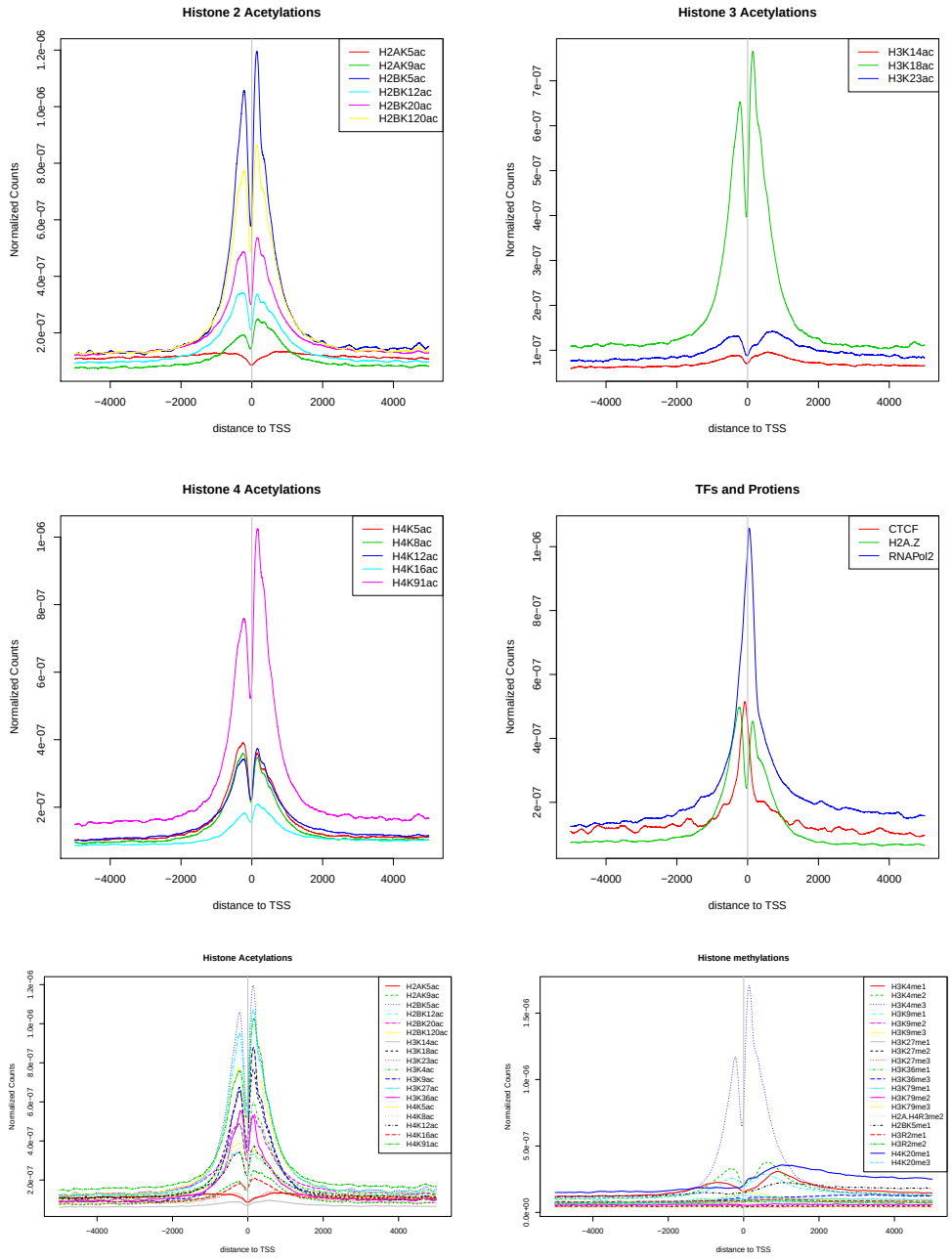

Figure 3.5: Histone modifications profiles at TSS

We now show profiles of histone modifications at the RNA polymerase II sites.

H3K4 modifications: H3K4me3 was most elevated at RNAPol2 binding sites but did not showed +1 characteristic nucleosome peak that one ob-

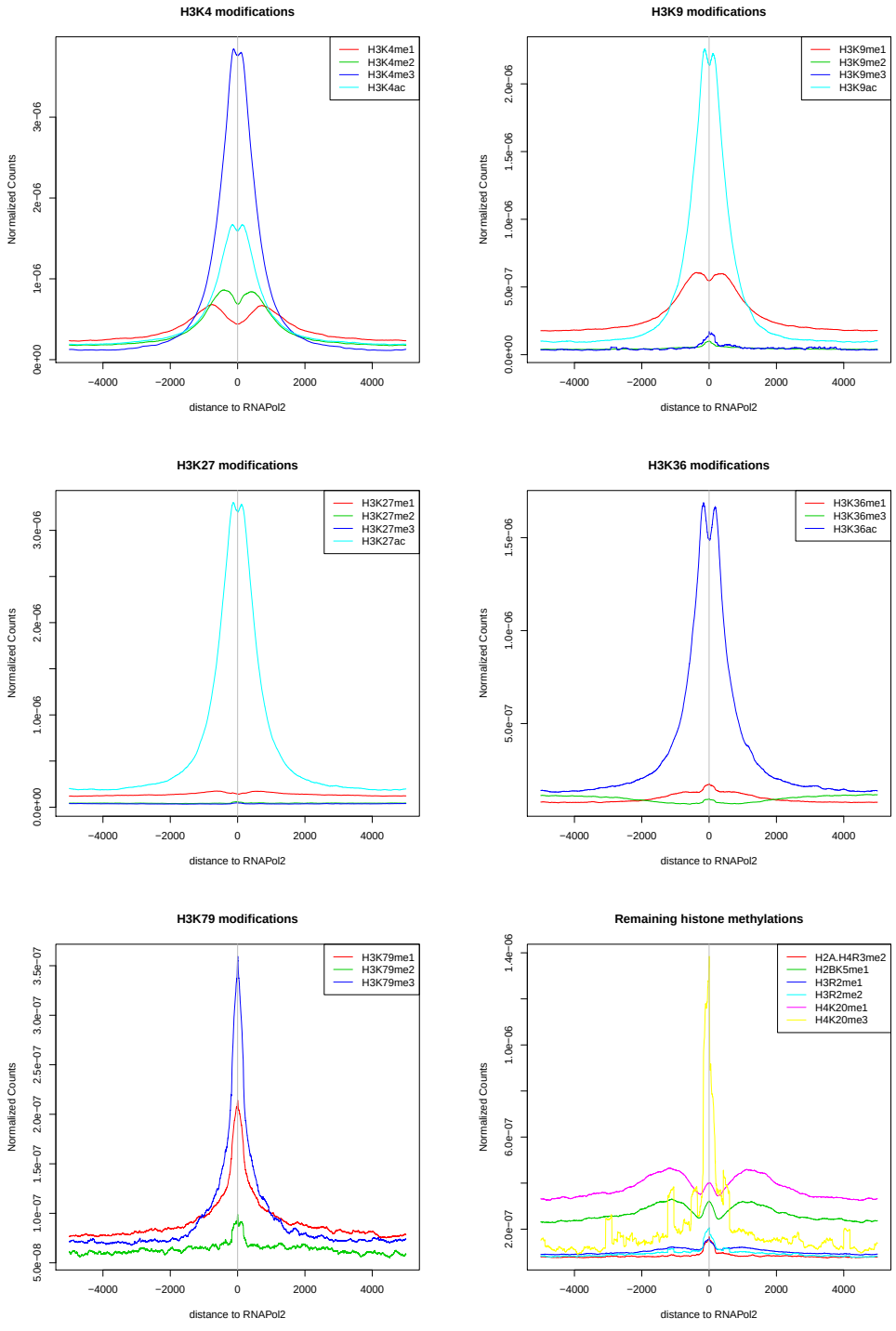

Figure 3.6: Histone modifications profiles at RNAPol2

serves around TSS. Infact most of the modifications showed symmetrical ±1 nucleosome peaks around RNAPol2 binding sites. H3K4ac had a symmetrical profiles around RNAPol2. H3K4me2 also showed some elevation around RNAPol2 binding sites. Like before at TSS, H3K4me1 showed a broader gap at RNAPol2 binding sites

H3K9 modifications: H3K9ac showed most enrichment, followed by H3K9me1, which had rather unusual ±1 nucleosome peaks. H3K9me2 and H3K9me3 showed almost zero enrichment at RNAPol2 binding sites.

H3K27 modifications: Of all modifications at H3K27 site, only H3K27ac showed high enrichment, while others showed almost no elevation.

H3K36 modifications: H3K36ac showed maximum enrichment at RNAPol2 binding sites. H3K36me1 and H3K36me3 showed slight elevation at RNAPol2, with H3K36me3 showing a elevation downstream of RNAPol2 binding sites.

H3K79 modifications Of all methylations at H3K79, H3K79me3 was most elevated, followed by H3K79me1. H3K79me2 showed slight enriched at RNApol2 binding positions.

Remaining Histone methylations: H4K20me3 was most interesting of remaining histone methylations. H4K20me3 showed step like peaks, both downstream and upstream of RNAPol2 binding sites. H4K20me1 and H2BK5me1 shared similar profiles but with different elevations. Rest of the modifications showed slight elevation at RNAPol2 binding positions.

H2 Acetylations: Except H2AK5ac and H2AK9ac, all other H2AK and H2BK acetylations had similar profiles with varying degrees of elevation. As seen previously H2AK5ac showed a sharp dip, while H2AK9ac showed a horizontal peak at RNAPol2 binding positions.

H3 Acetylations: H3K18ac like H3K27ac was highly at elevated either side of RNAPol2 binding sites. H3K14ac showed almost no enrichment, while H3K23ac showed broad peaks either side, which were very slightly enriched.

H4 Acetylations: All except H4K16ac, showed similar profiles, with varying degrees of enrichment around RNAPol2 binding sites. H4K16ac had a constant peak, showing no obvious dip at RNAPol2 binding sites.

TFs and Proteins at TSS H2A.Z had a equal bimodal peak around RNAPol2 binding sites. Surprisingly, CTCF was also enriched at RNAPol2 binding sites

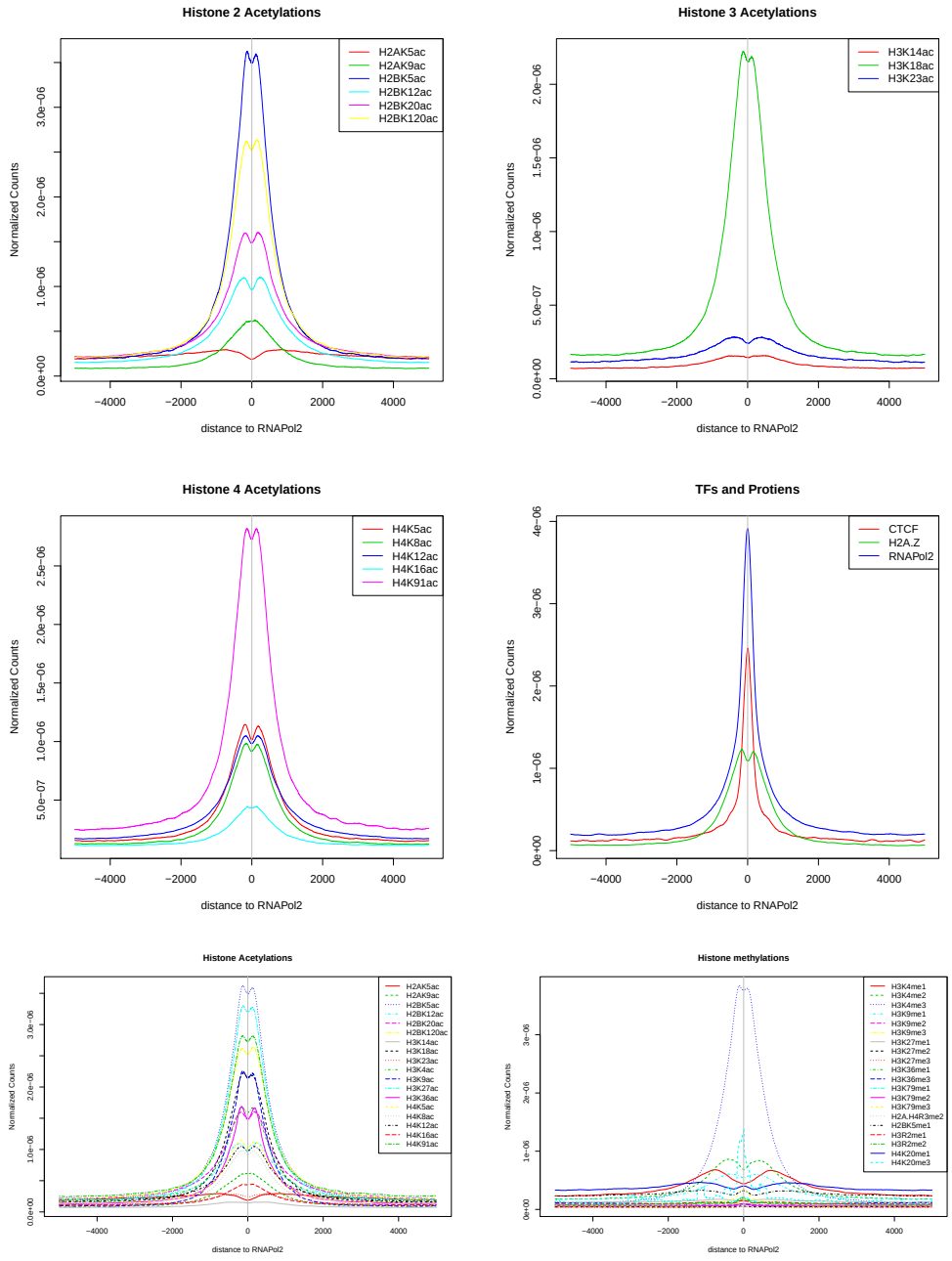

Figure 3.7: Histone modifications profiles at RNAPol2

3.5 Histone modifications at enhancers

Enhancers are defined by a function instead of a specific sequence or by some regulatory element, and in function are similar to proximal promoter elements.

Enhancers are short sequences which, when placed upstream/downstream of a promoter can activate them.

"p300" is one of the common bridges between a lot transcription factors and promoters. It has been found that there has been a high correlation between p300 binding sites and neighbouring gene expression. p300 is thought to increase gene expression in many ways. Some of them are

- opening chromatin at promoters via histone acetyltransferase activity
- recruiting RNA polymerase II and basal transcriptional machinery to the promoters

We used p300 binding sites to identify enhancer locations. We got p300 data from [27]. We removed histone 3 lysine 4 trimethylation(H3K4me3) marks that close to the p300 binding sites, to find potential enhancers. We now show profiles of histone modifications at these enhancers.

H3K4 modifications: All four states of H3K4 showed some elevation at enhancer positions. H3K4me1 was most elevated followed by H3K4me2 and H3K4ac. H3K4me3 was least elevated. This result is consistent with previous finding of histone modifications at enhancer locations [28].

H3K9 modifications: H3K9me1 was most elevated of all the states of H3K9. This was followed by H3K9ac. H3K9me3 showed a sharp peak at the enhancer locations, while H3K9me2 was least elevated of all the modifications.

H3K27 modifications: H3K27ac was most enriched with at the enhancer locations and this was consistent with finding in [28, 29]. Other marks also showed some elevation but to lesser degrees. **H3K36 modifications:** H3K36ac showed maximum elevation at enhancers and had higher ± 1 nucleosome peaks, followed by H3K36me1 and H3K36me3, which also showed enrichment at the enhancer positions.

H3K79 modifications All three state at H3K79 had similar profile. H3K79me3 showed maximum elevation followed by H3K79me1 and H3K79me2. None of them showed distinct ± 1 peaks.

Remaining Histone methylations: H4K20me3 was most enriched at enhancers, showing unusual peaks both upstream and downstream of the enhancer locations. H2A.H4R3me2, H3R2me1, H3R2me2 showed similar profiles, while H2BK5me1 and H4K20me1 showed similar profiles.

H2 Acetylations: H2BK5ac, H2BK12ac, H2BK20ac, H2BK120 at Histone 2 showed enrichment around the enhancer positions to varying degrees, with distinct ± 1. H2AK5ac showed sharp dip at enhancer positions, while H2AKac showed minor enrichment.
H3 Acetylations: H3K18ac was most enriched of remaining H3 acetylations, with prominent ± 1, but the orientation of $\pm$ peaks was opposite as compared to H3K27ac. H3K18ac and H3K23ac showed some enrichment at enhancer locations
H4 Acetylations: All H4 acetylations had more or less similar patterns except H4K16ac which showed some enrichment at enhancer positions but no enrichment at ± 1 nucleosome peaks
TFs and Proteins at TSS Surprisingly, CTCF showed sharp peak at enhancer peaks and was more enriched as compared to histone modifications.

3.6 Histone modifications at CTCF

Insulators are again defined by a function instead of a sequence or a particular region in the genome. Some of the many important functions of Insulator are

- Insulator binding sites act as boundaries between euchromatin and heterochromatin. Insulators are also known to block the interaction between enhancers and promoters[30]
- Insulator binding sites act as anchors for nucleosome positioning to align various genomic signals[31]

CC**CTC**-binding **F**actor is a insulator binding protein and a transcription factor that is encoded by CTCF gene [30]. There are approximately 15,000 insulator binding CTCF sites and distribution of these sites is found to be in high correlation with distribution of gene expression [32, 31].

We believe that a careful analysis of hsitone modification patters at insulator binding positions will to correctly predict its role in enhancer-promoter blocking. We now show profiles various histone modification at CTCF binding locations.

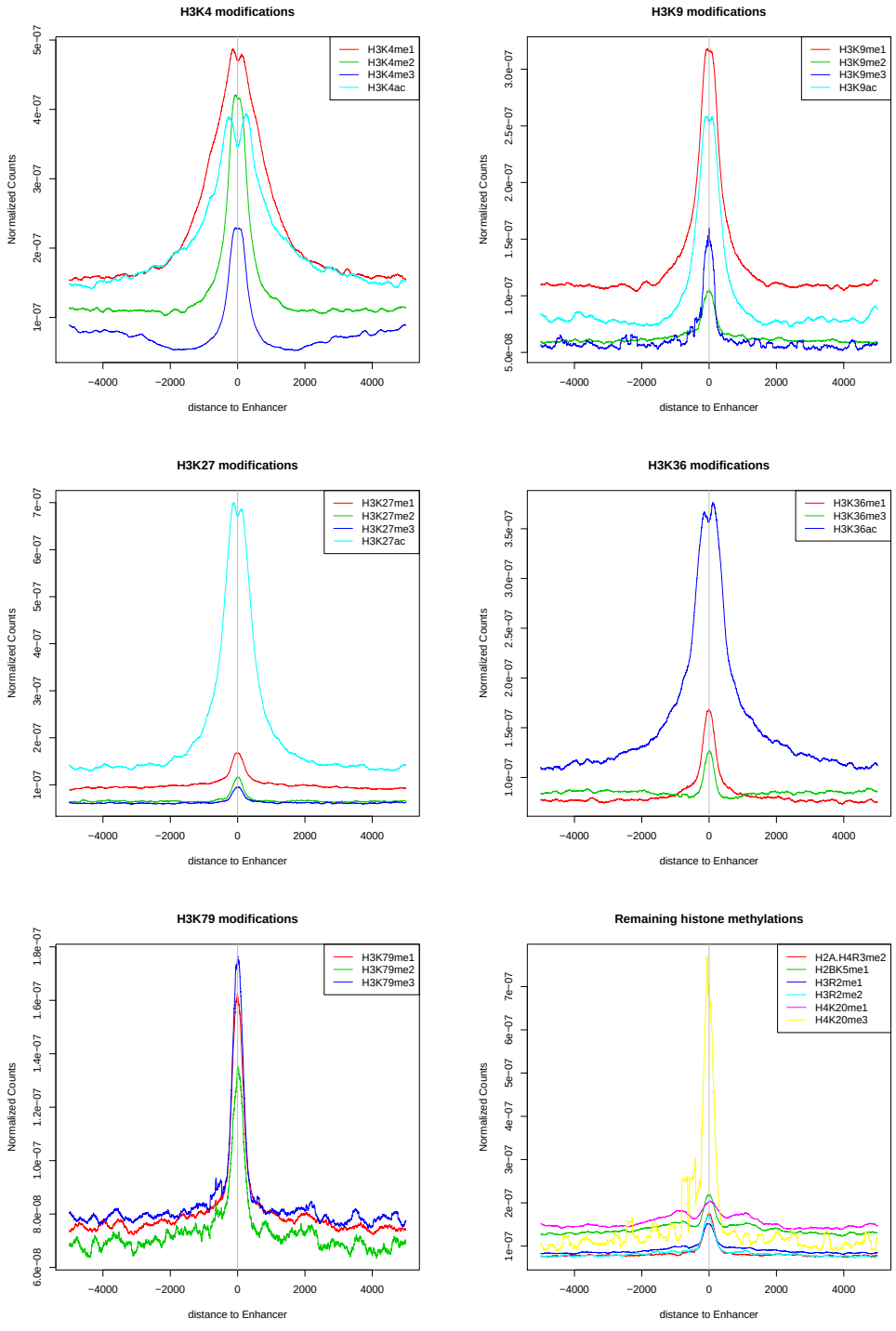

Figure 3.8: Histone modifications profiles at Enhancers

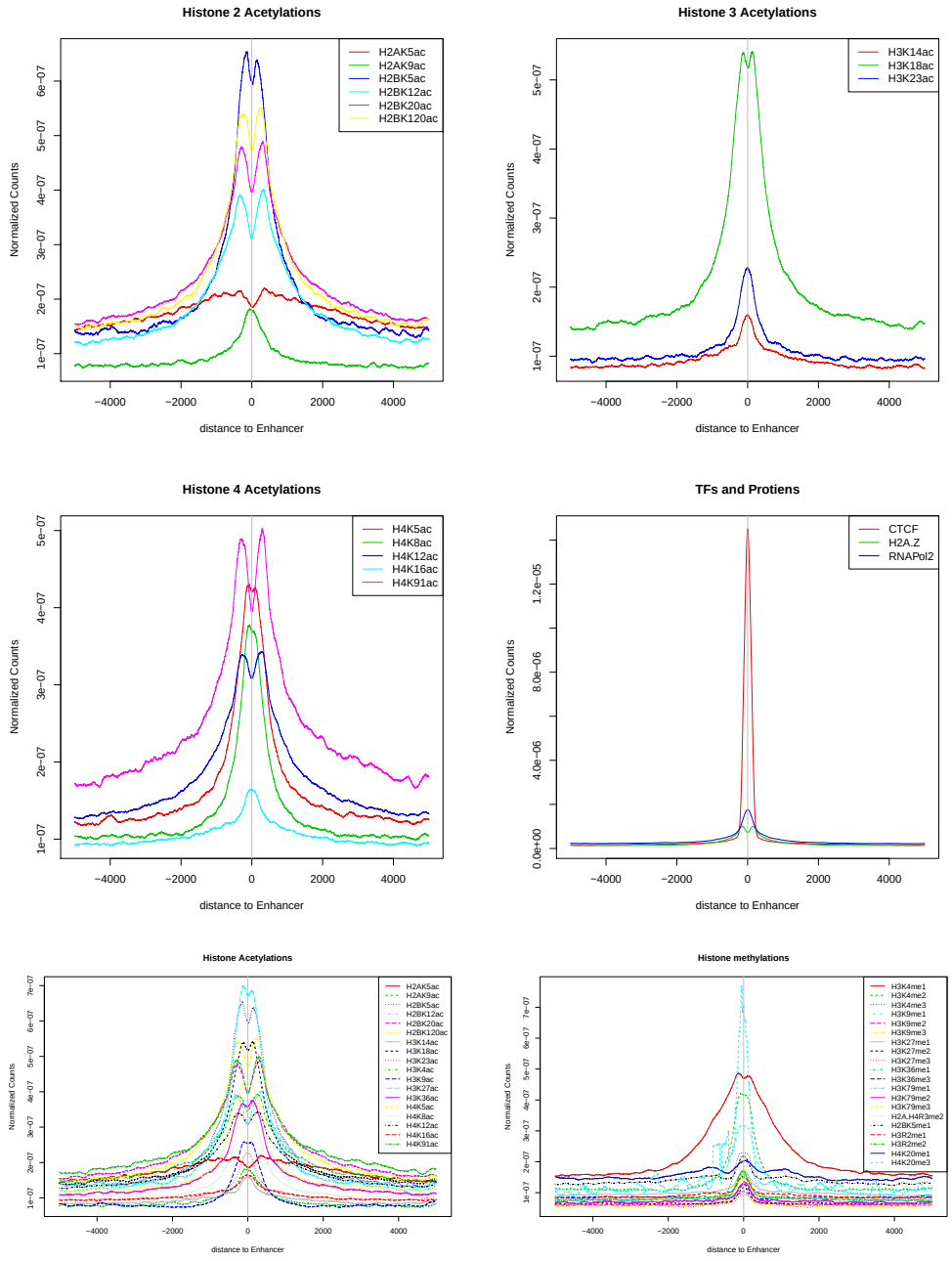

Figure 3.9: Histone modifications profiles at Enhancers

H3K4 modifications: All states of H3K4 showed near symmetrical ±1 nucleosomal peaks around CTCF binding sites. H3K4me3 was most elevated with other modifications more or less equally elevated.

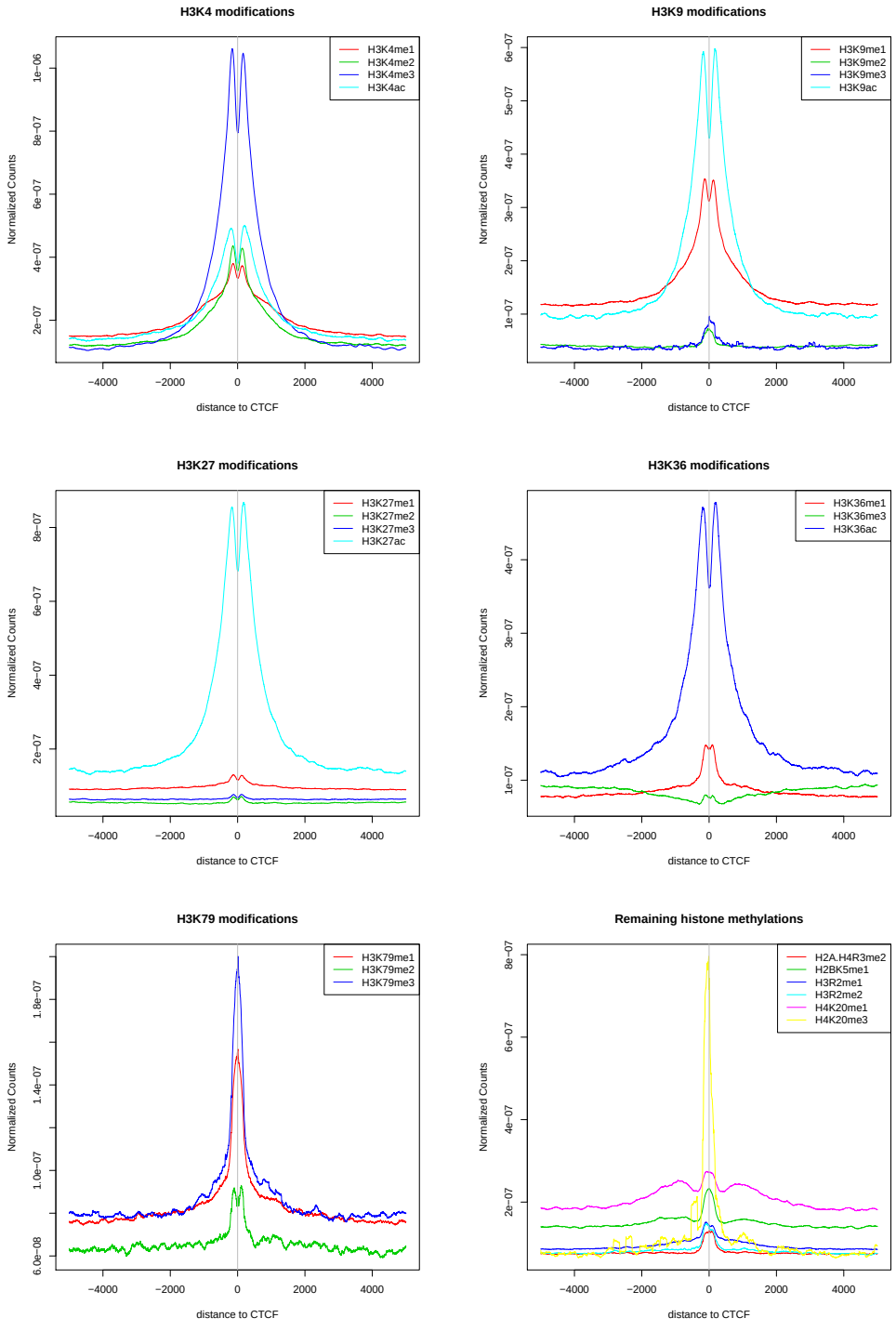

Figure 3.10: Histone modifications profiles at CTCF

H3K9 modifications: Of all states at H3K9, H3K9ac was most elevated with ±1 peaks, followed by H3k9me1. The other two states showed no elevation.

H3K27 modifications: Surprisingly, only H3K27ac showed elevation at CTCF binding sites, which known to have association with transcriptional activation. All other three states showed some elevation around CTCF but to a lesser degree.

H3K36 modifications: H3K36ac was most elevated of H3K36 states. H3K36me3 like before showed increasing elevation both upstream and downstream of CTCF binding sites. H3K36me1 showed moderate elevation

H3K79 modifications Two H3K79 states were most strange of all the histone modifications. H3K79me1 and H3K79me3 showed elevation at the CTCF binding sites, while H3K79me2 showed usual ±1 peaks but was very less enriched.

Remaining Histone methylations: Like in enhancers, H4K20me3 was again enriched at CTCF positions, showing peculiar peaks both upstream and downstream of the CTCF binding positions. H2A.H4R3me2, H3R2me1, H3R2me2 showed similar profiles, while H2BK5me1 and H4K20me1 showed similar profiles.

H2 Acetylations: H2BK5ac, H2BK12ac, H2BK20ac, H2BK120 at Histone 2 showed enrichment around the CTCF positions to varying degrees, with distinct ±1. H2AK5ac showed sharp dip at CTCF positions, while H2AKac showed more enrichment than it had at enhancer locations.

H3 Acetylations: H3K18ac was most enriched of remaining H3 acetylations, with prominent ±1 peaks. H3K18ac and H3K23ac showed some enrichment at CTCF locations with ±1 peaks

H4 Acetylations: Of all H4 acetylations H4K91ac was most enriched, which is similar to its profile at TSS. H4K5ac, H4K8ac and H4K12ac had all similar profiles as they had at TSS. H4K16ac also showed some enriched around CTCF binding positions. Overall, H4 acetylations profiles at CTCF were similar to TSS.

TFs and Proteins at TSS As expected CTCF was most enriched at its own binding position. Others showed almost zero enrichment.

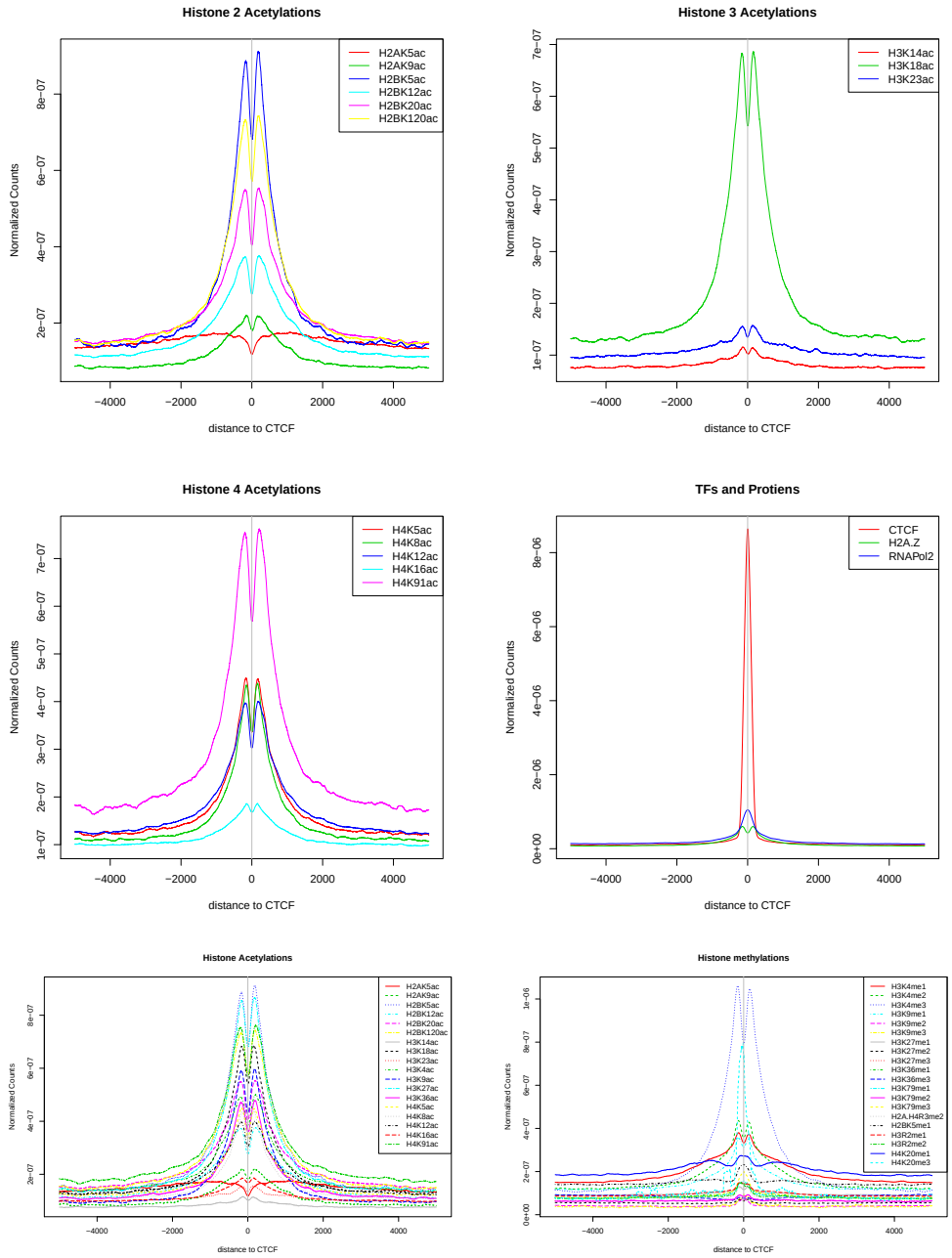

Figure 3.11: Histone modifications profiles at CTCF

Chapter 4

Learning combinatorial patterns of histone modifications under Boolean network model

> *One, two, Freddy's coming for you*
> *Three, four, better lock your door*
> *Five, six, grab your crucifix*
> *Seven, eight, gotta stay up late*
> *Nine, ten, never sleep again*
>
> — A Nightmare on Elm Street
> *A poem variant- One, two, Buckle.Z*

Boolean networks at their core are a bit like simple counting. Most of the algorithms in this chapter are based on simple counting, whether its counting transition from one state to another in best-fit algorithm, or the occurrence of a boolean state in the data in network selection algorithm.

In this chapter, we address the problem of inferring the structure of histone networks under boolean network model. Histone modification data can be noisy or inconsistent due to measurement errors at immunoprecipitation step. Best-fit extension problem can incorporate such inconsistencies and using it we can establish a histone network that would make very few misclassifications.

After robustly searching the first three features with best-fit algorithm , we then add and remove one feature at time using floating backward-forward algorithm based on hold-out cross validation error . Finally, we identify most prominent histone modification network using network select algorithm.

4.1 Background and definitions

A boolean network consist of a set Boolean variables, whose state is determined by other variables in the network. In the present case, the variables are histone modifications, and we want to learn the state of a histone modification given the other modifications.

For consistency of notation with previous work, we will follow the same notation as in [33, 34]. We represent the boolean network of histone modification by $G(C, F)$, where $C = (c_1, c_2, ..., c_n)$ represents a set nodes of histone modifications and $F = (f_1, f_2, ..., f_n)$ represents the boolean functions. A boolean function $f_i(c_{i_1}, c_{i_2}, ..., c_{i_k})$ with k histone modification assigned to modification c_i. This also allows the unnecessary chromatin modifications in each function to be fictitious.

For example, for a function f, a modification c_i is fictitious if

$$f(c_1, ..., c_{i-1}, 0, c_{i+1}, ..., c_n) =$$

$$f(c_1, ..., c_{i-1}, 1, c_{i+1}, ..., c_n)$$

$\forall$ values of $c_1, c_2, ..., c_{i-1}, c_{i+1}, ..., c_n$. A variable that is not fictitious is called essential. Each node c_i represents the state of modification i, where $c_i = 1$ means modification is present in the nucleosome and $c_i = 0$ means modification is not present in the nucleosome.

4.2 Best fit algorithm

In real situations, histone modification measurements can be noisy due to various factors- for example, due to bad alignment during the mapping stage, or due to non-validated antibodies or due to some impurities. All these can result

in noise, which can lead to inconsistent assignment of histones to nucleosomes. Best-fit extension Problem is a learning strategy that can incorporate such inconsistencies [34]. The aim of Best-fit extension problem is to establish a network structure that would make as few misclassifications as possible. A detailed description of Best-fit extension problem is provided in [35].
A partially defined Boolean function(pdBf) is defined by a pair of sets

$$T, F \subseteq \{0,1\}^n$$

A function f is called an extension of pdBf(T,F) if

$$T \subseteq T(f) \ \& \ F \subseteq F(f) \tag{4.1}$$

where

$$T(f) = \{x \in \{0,1\}^n : f(x) = 1\} \tag{4.2}$$

$$F(f) = \{x \in \{0,1\}^n : f(x) = 0\} \tag{4.3}$$

i.e. T is a set of true vectors and F is a set of false vectors. Supposing we are also given positive weights $w(x)$ for all vectors $x \in T \cup F$. We define $w(S)$ for a subset $S \subseteq T \cup F$

$$w(S) = \sum_{x \in S} w(x) \tag{4.4}$$

Now we define the error size of function f under Best-Fit extension problem is defined as

$$\varepsilon(f) = w(T \cap F(f)) + w(F \cap T(f)) \tag{4.5}$$

If the weights are one i.e. $w(x) = 1$ for all $x \in T \cup F$, then the error is just the number of misclassifications. In our analysis of histone modifications, we had made the above assumption. Now our goal is output subsets T^* and F^* such that

$$T^* \cap F^* = \emptyset \ \& \ T^* \cup F^* = T \cup F \tag{4.6}$$

For the above equation, pdBf(T^*, F^*) has an extension in some class of functions so that $w(T \cap F(f)) + w(F \cap T(f))$ is minimum. Thus any extension of pdBf will have minimum error size [34].

4.3 Hold-out cross validation

To estimate the performance of the boolean network model, we use hold-out cross-validation(also called simple cross validation). In hold-out cross validation, we divide the data into training(70% of the data) and test(30% of the data) sets. The inputs are X (input features), Y(target features), w (weight variable), k(number of selected features). It returns the least error on the hold out cross validation set.

Algorithm *holdOutValidation*
Input: X,Y,w,k
Output: Ehat

1. Split the data into training D_{train} and test D_{test} data.
2. ($*$ Here, D_{test} is called hold-out cross validation set $*$)
3. Train each model on D_{train} using Best-Fit, to get a set hypothesis function
4. Select and output the hypothesis function that gives least error on hold out cross validation set D_{test}.

4.4 Selecting histone modifications

In machine learning, feature selection or variable selection is a technique for selecting a minimum subset of features required to build a robust learning model. For our problem, we applied floating forward-backward feature selection as proposed in [36]. After, robustly searching a combination of 3 variables that would gives us least error for the target histone modification, we then, from remaining set of features, added a feature and removed a feature, one at a time, hence getting a list of 38 errors after each stage. We choose the combination that gave us the least error and we repeated the process till the error decreased.

Algorithm *floatingFeature*

```
Input: targetVariable
Output: Set of features
1.  k ← 3
2.  (oldError , features) ← holdOutValidation(k features)
3.  featurePool ← totalFeatures - features
4.  while (1)
5.          for i in features
6.              do
7.                  featureDec ← features - features(i)
8.          for i in featurePool
9.              do
10.                 featureInc ← features + featurePool(i)
11.         newErrorInc ← holdOutValidation(featureInc, k + 1 features)
12.         newErrorDec ← holdOutValidation(featureDec, k − 1 features)
13.         newError ← min(newErrorInc, newErrorDec)
14.         if newError < oldError
15.                  if newError ∈ newErrorInc
16.                          add feature(newErrorInc) to features
17.                          remove feature(newErrorInc) from featurePool
18.                          k ← k + 1
19.                    else  remove feature(newErrorDec) from features
20.                          add feature(newErrorDec) to featurePool
21.                          k ← k − 1
22.                   oldError ← newError
23.             else
24.                     break
25.   if size(features) ==1
26.             break
27.   if sizeOf(featurePool) == 0
28.             break
29.return (features, newError)
```

4.5 Selecting histone modification networks

In network selection algorithm, we study the possible inhibition or assistance that selected features might have on the target histone modification. After we got a list of features from features selection algorithm, we then construct a truth table for selected features and identified the most occurring combination, and noted the corresponding value of the target feature. Suppose we identified 3 features, P1, P2, P3 and suppose our target feature is P0. Now we scan each 2^3 boolean combinations through the data-set and count the number of times the target variable was zero or one. Now we selected the most occurring value of the target variable and the corresponding boolean combination. In the table below, we would select C_0 equal to 60000 and the corresponding network combination that would be P1(0), P1(1), P2(1) i.e the fourth row in table below. This is basically what we have done in network select algorithm.

P1	**P2**	**P3**	C_0	C_1
0	0	0	10	45
0	0	1	200	21
0	1	0	54	480
0	1	1	60000	50000
1	0	0	3	1000
1	0	1	19000	8000
1	1	0	0	3
1	1	1	4000	9000

Algorithm *networkSelect*
Input: selected features, target feature, input data
Output: target value, target pattern
1. truthMat ← truth table(no. of selectedFeatures)
2. inputData ← input data(selectedFeatures)
3. **for** i in sizeof(inputData)
4. locBinMat ← position of inputData(i) in truthmat
5. (∗ locBinMat ranges from 1 to pow(2,size(selectedFeatures))-1 ∗)

```
6.          if inputData(i,targetVariable) = 0
7.             then increment countMatrix(locBinMat,1)
8.             else  increment countMatrix(locBinMat,2)
9.  maxCount ← maximum value in CountMatrix
10. maxPos ← find(maxCount in CountMatrix)
11. maxPosRow ← floor((find(maxCount in CountMatrix)/2)
12. targetPattern ← truthMat(maxPosRow)
13. if maxPos > pow(2,sizeof(selectedFeatures))
14.    then targetValue ← 1
15.    else  targetValue ← 0
16. return (targetValue, targetPattern)
```

4.6 Results

In this section we discuss results for some of the important histone modifications for which we got least error, and also compare them with correlation and clustering results.

One of the highly studied histone modification is H3K27ac. It is involved in chromatin decondensation, and we also found it to be present at most regulatory regions (refer chapter 3). The marks that were associated with H3K27ac in both clustering and boolean analysis were H3K18ac, H2BK120ac, H4K12ac and H3K36ac. In boolean analysis, we found H3K4me3 mark to be closely associated with H3K27ac mark, but was shown in a different cluster in the clustering result 2.6. Also we found that H3K27me3 was absent when the acetylation mark at the same site was present. In addition, we found H3K27me2, H4K5ac, H4K8ac and H3K9ac to be closely associated with H3K27ac. The cross validation error that we got was 0.06.

One of the other important mark is H3K4me3. It is a characteristic mark of promoters, and is involved in a lot of transcription related processes. Some of the recent studies [39, 40] have shown that H3K4me3 complex plays a critical role in regulating lifespan across many generations. In the clustering results we found H3K9ac, H2BK5ac, H2AZ, H4K8ac and H4K5ac to be in the same cluster as H3K4me3. While boolean analysis showed presence of H2AZ,

H2BK5me1 and H3K9me3. Also there was absence of H3K27ac. This shows that H3K4me3 is required for presence H3K27ac, but vice versa might not hold true.

Overall, the clustering results and boolean analysis differed for H3K4me3 mark, but were consistent for H3K27ac. This also gives us incentive to experiment with other feature selection algorithms and clustering methods.

Following are results for 39 histone modifications, and the corresponding histone marks that showed the dependency. The last column gives the corresponding error.

H2A.H4R3me2	H3K9me2	H3R2me2	H4K20me3	H3K27me2	H3K79me3				Error
0	0	0	0	0	0				0.011
H2AZ	H3K4me3	H4K20me1	H3K18ac	H3K9me1	H3K4me1	H3K27ac	H3K27me3	H3K4me2	Error
0	0	0	0	0	1	1	0	0	0.063
H2BK5me1	H3K4me1	H3K4me3	H3K9me1	H3K27me1	H4K20me1	H4K20me3	H3K18ac	H3K79me3	Error
1	1	1	0	1	0	0	1	1	0.072
H3K4me1	H3K4me2	H3K4me3	H3K9me1	H3K27me1	H3K36me3	H3K9ac	H3K23ac	H2AK5ac	Error
0	0	1	1	1	0	0	1	1	0.086
H3K4me2	H3K4me1	H3K4me3	H3K9me1	H3K23ac	H4K8ac	H4K91ac	H4K20me3	H3K27me1	Error
1	1	1	1	1	0	1	0	1	0.084
H3K4me3	H2AZ	H2BK5me1	H3K4me2	H3K9me1	H3K9me3	H3K27me1	H3K9ac	H3K27ac	Error
0	0	0	1	1	0	1	1	1	0.059
H3K9me1	H3K4me1	H3K4me2	H3K4me3	H3K79me3	H4K20me1	H4K20me3	H2BK120ac	H3K36ac	Error
1	1	1	1	0	0	0	1	1	0.087
H3K9me2	H2A.H4*	H3K27me2	H3K79me1	H3R2me2	H4K20me3	H3K36me1			Error
0	0	0	0	0	0	0			0.012
H3K9me3	H2AZ	H3K36me3	H4K20me3	H3K27ac	H2A.H4*	H3K9me2	H4K5ac		Error
0	0	1	0	0	1	0	0		0.006
H3K27me1	H2BK5me1	H3K9me1	H3K23ac	H3K36me3	H3K4me2	H3K27me2	H3K4me1		Error
0	0	1	1	0	0	1	0		0.095
H3K27me2	H3K9me2	H3K79me1	H4K20me3	H4K20me1	H3K27me3	H3K27me1	H3K36me3		Error
0	0	0	0	0	0	1	1		0.005
H3K27me3	H2A.H4*	H3K4me3	H3K9me3	H3K79me1	H3K79me2	H2AK5ac			Error
0	0	0	0	1	1	1			0.004
H3K36me1	H2BK5me1	H3K9me2	H3K79me1	H3K79me2	H3R2me1	H2BK12ac			Error
0	0	0	1	1	1	1			0.031
H3K36me3	H2AZ	H3K4me1	H3K9me3	H3K27me1	H3K79me3	H4K20me3	H3K18ac	H3K23ac	Error
0	0	0	0	1	1	1	1	1	0.039

H3K79me1	H3K9me2	H3K79me3	H3R2me1	H3R2me2	H2AK9ac	H3K4me1	H2AZ		Errror
0	0	0	0	0	0	0	0		0.015
H3K79me2	H3K9me3	H3K27me3	H3K14ac	H4K16ac	H3R2me2	H2BK12ac			Error
0	0	0	0	0	0	0			0.004
H3K79me3	H3K27me3	H3K36me3	H4K20me3	H3K9me1	H3K9me3	H3K79me2	H4K20me1		Error
0	0	0	0	0	1	0	0		0.020
H3R2me1	H2A.H4*	H3K36me1	H3K79me3	H3K27me3	H2BK5me1	H3K79me1	H3K9me1	H2BK120ac	Error
0	0	0	0	0	0	0	0	1	0.031
H3R2me2	H2A.H4*	H3K9me2	H4K20me3	H2AZ	H3K79me1	H3R2me1			Error
0	0	0	0	0	0	1			0.022
H4K20me1	H2AZ	H2BK5me1	H4K5ac	H3K9me1	H3K18ac	H2AK5ac	H3K27ac	H4K20me3	Error
1	1	0	1	0	0	1	1	1	0.075
H4K20me3	H2A.H4*	H3K4me1	H3K9me1	H3K9me3	H3K79me3	H4K20me1	H4K91ac		Error
0	0	0	1	1	0	0	1		0.013
H3K4ac	H2BK12ac	H2BK20ac	H2BK120ac	H4K91ac	H4K12ac	H3K18ac	H3K27ac	H2AK5ac	Error
1	1	1	1	1	1	1	1	1	0.071
H3K9ac	H3K4me3	H3K27ac	H2BK120ac	H3K4me1	H3K18ac	H2BK5ac	H2AZ	H3K4ac	Error
0	0	0	0	0	0	1	1	1	0.058
H3K14ac	H3K9me3	H3K79me2	H3R2me1	H3R2me2	H4K16ac	H2AK5ac			Error
0	0	0	0	0	0	0			0.028
H3K18ac	H2AZ	H3K4me3	H4K20me1	H3K4ac	H3K27ac	H3K36ac	H4K91ac	H2BK20ac	Error
0	0	0	0	0	1	1	1	1	0.063
H3K23ac	H3K4me2	H3K9me3	H3K79me2	H2BK5ac	H2BK5me1	H4K20me3			Error
0	0	1	0	0	0	0			0.060
H3K27ac	H3K18ac	H2BK5ac	H2BK120ac	H3K4me3	H4K5ac	H3K9ac	H4K8ac	H3K27me3	Error
1	1	1	1	1	1	1	1	0	0.064
H3K36ac	H3K18ac	H2BK20ac	H2BK120ac	H3K4ac	H2BK12ac	H3K9ac	H2AZ	H3K27ac	Error
0	0	0	0	0	0	0	0	0	0.063

H4K5ac	H3K27ac	H3K36ac	H4K8ac	H2BK12ac	H3K23ac	H4K12ac	H2AK5ac	H3K18ac	Error
0	0	0	0	0	0	0	0	0	0.077
H4K8ac	H2AZ	H3K9ac	H4K5ac	H3K36ac	H4K12ac	H3K23ac	H2AK9ac	H3K18ac	Error
0	0	0	0	0	0	0	1	1	0.071
H4K12ac	H3K9me3	H3K4ac	H2BK20ac	H4K20me1	H2BK12ac	H3K36me3			Error
0	0	0	0	0	0	1			0.040
H4K16ac	H3K9me3	H3K4ac	H3K9ac	H3K36ac	H4K5ac	H4K8ac	H4K12ac	H4K91ac	Error
0	0	0	0	0	0	0	0	1	0.076
H4K91ac	H3K9me3	H3K4ac	H3K18ac	H3K27ac	H3K36ac	H4K16ac	H2AK5ac	H2BK120ac	Error
1	1	0	1	1	1	1	1	1	0.081
H2AK5ac	H3K4me3	H2BK12ac	H2BK20ac	H4K91ac	H3K27me2	H3R2me1	H2BK5me1		Error
0	0	0	0	0	0	0	0		0.061
H2AK9ac	H3K4me2	H3K9me3	H3K36me1	H3R2me2	H3K14ac	H4K91ac	H3K9me1		Error
0	0	1	0	0	0	0	1		0.040
H2BK5ac	H3K18ac	H3K27ac	H2BK120ac	H3K9ac	H4K12ac	H3K27me2	H4K8ac	H3K9me1	Error
0	0	0	0	0	0	0	0	0	0.070
H2BK12ac	H3K4me1	H3K4ac	H3K36ac	H4K5ac	H4K12ac	H4K91ac	H2AK5ac	H2BK120ac	Error
0	0	0	1	1	1	1	1	1	0.068
H2BK20ac	H3K18ac	H4K12ac	H4K91ac	H3K4me3	H2AK5ac	H3K79me1	H3K79me2		Error
1	1	1	1	1	0	1	1		0.037
H2BK120ac	H3K4ac	H3K18ac	H4K12ac	H4K91ac	H2BK5ac	H2BK12ac	H2A.H4*		Error
1	1	1	1	1	1	1	0		0.066

Table 4.1: The binary combinatorial histone code: results for 39 histone modifications, followed by histone marks that show best dependency. The corresponding row gives the most frequent occurring combination of the selected histone modifications, and the last column gives the corresponding error.

Chapter 5

Summary and future work

> *I should have more faith, I ought to know by this time that when a fact appears opposed to a long train of deductions it invariably proves to be capable of bearing some other interpretation.*
>
> — Sherlock Holmes
> *What is an activating mark ?*

5.1 Summary and conclusions

To summarize, we predicted nucleosomes by combining histone acetylations and methylations data from [13, 14]. We then assigned the histone modifications to the predicted nucleosomes. A modification was set to 1 if it was present in the nucleosome else it was set to zero. After this, we robustly searched combination of three modifications which would give least error, then added or removed one modification at a time till the error decreased. Once we had the list of features, we then constructed a Truth Table(TT), for the selected features and counted each of them through the dataset, to see which combination occurs most and also noted the corresponding TargetValue(which will be either 0 or 1). Thus finally we had a list for each modification showing which other modifications were on or off when this modification was present or absent.

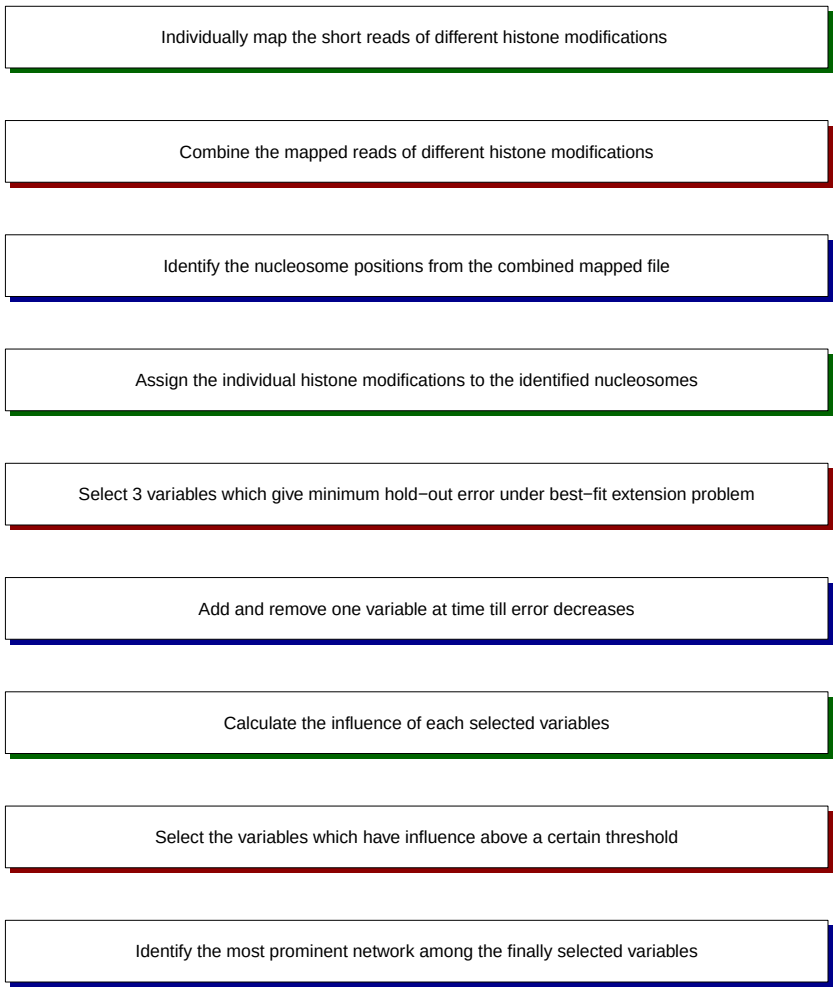

Figure 5.1: The overall workflow of thesis

5.2 Future work

In future work, we wish to do the following

1. Extend the same analysis to other human cells, to check the quality of our predictions
2. Extend the analysis to other organisms like yeast and flies, and observe what difference we get
3. Try out other machine learning methods like regression based models

and compare the results with boolean network model

After we have a model, then we plan to extend the analysis to other regulatory regions like enhancer, insulators and learn how modifications vary as one goes from one regulatory region to another. After this we aim define new annotations for various regulatory based on our newly found histone modifications dependencies. Currently, only a limited number of histone modifications are used to characterize a regulatory region. Since nucleosomes are characteristic at a regulatory, currently available software's fail to identify good quality nucleosomes genome-wide. Finally, we want come up with a nucleosome prediction software that would predict nucleosomes characteristic of various regulatory elements or regions.

Bibliography

> *Stand on the shoulders of giants*
>
> — Google Scholar
>
> *Yet another variant(Newton's original quote)*

[1] G. Felsenfeld and M. Groudine. Controlling the double helix. *Nature*, 421(6921):448–453, 2003.

[2] CD Allis, T. Jenuwein, and D. Reinberg. Epigenetics. 2007.

[3] R.D. Kornberg and Y. Lorch. Twenty-five years of the nucleosome, review fundamental particle of the eukaryote chromosome. *Cell*, 98:285–294, 1999.

[4] `http://en.wikipedia.org/wiki/Histone`.

[5] K. Van Holde and J. Zlatanova. What determines the folding of the chromatin fiber? *Proceedings of the National Academy of Sciences*, 93(20):10548, 1996.

[6] A. Lusser and J.T. Kadonaga. Strategies for the reconstitution of chromatin. *Nature Methods*, 1(1):19–26, 2004.

[7] M. Rodríguez-Paredes and M. Esteller. Cancer epigenetics reaches mainstream oncology. *Nature Medicine*, pages 330–339, 2011.

[8] http://www.nobelprize.org/educational/medicine/dna/a/transcription/acetylation.html.

[9] R.J. Klose and Y. Zhang. Regulation of histone methylation by demethylimination and demethylation. *Nature Reviews Molecular Cell Biology*, 8(4):307–318, 2007.

[10] T. Kouzarides. Chromatin modifications and their function. *Cell*, 128(4):693–705, 2007.

[11] E.R. Mardis. Chip-seq: welcome to the new frontier. *Nature methods*, 4(8):613–613, 2007.

[12] B. Langmead, C. Trapnell, M. Pop, and S.L. Salzberg. Ultrafast and memory-efficient alignment of short dna sequences to the human genome. *Genome Biol*, 10(3):R25, 2009.

[13] A. Barski, S. Cuddapah, K. Cui, T.Y. Roh, D.E. Schones, Z. Wang, G. Wei, I. Chepelev, and K. Zhao. High-resolution profiling of histone methylations in the human genome. *Cell*, 129(4):823–837, 2007.

[14] Z. Wang, C. Zang, J.A. Rosenfeld, D.E. Schones, A. Barski, S. Cuddapah, K. Cui, T.Y. Roh, W. Peng, M.Q. Zhang, et al. Combinatorial patterns of histone acetylations and methylations in the human genome. *Nature genetics*, 40(7):897–903, 2008.

[15] Y. Zhang, H. Shin, J. Song, Y. Lei, and X.S. Liu. Identifying positioned nucleosomes with epigenetic marks in human from chip-seq. *BMC genomics*, 9(1):537, 2008.

[16] N.J. Sakabe and M.A. Nobrega. Genome-wide maps of transcription regulatory elements. *Wiley Interdisciplinary Reviews: Systems Biology and Medicine*, 2(4):422–437, 2010.

[17] N.D. Heintzman, R.K. Stuart, G. Hon, Y. Fu, C.W. Ching, R.D. Hawkins, L.O. Barrera, S. Van Calcar, C. Qu, K.A. Ching, et al. Distinct and predictive chromatin signatures of transcriptional promoters and enhancers in the human genome. *Nature genetics*, 39(3):311–318, 2007.

[18] N.D. Heintzman, G.C. Hon, R.D. Hawkins, P. Kheradpour, A. Stark, L.F. Harp, Z. Ye, L.K. Lee, R.K. Stuart, C.W. Ching, et al. Histone modifications at human enhancers reflect global cell type-specific gene expression. *Nature*, 459(7243):108, 2009.

[19] H.H. He, C.A. Meyer, H. Shin, S.T. Bailey, G. Wei, Q. Wang, Y. Zhang, K. Xu, M. Ni, M. Lupien, et al. Nucleosome dynamics define transcriptional enhancers. *Nature genetics*, 42(4):343–347, 2010.

[20] O.J. Rando and H.Y. Chang. Genome-wide views of chromatin structure. *Annual review of biochemistry*, 78:245, 2009.

[21] R.D. Hawkins, G.C. Hon, and B. Ren. Next-generation genomics: an integrative approach. *Nat Rev Genet*, 11(7):476–486, 2010.

[22] M. Radman-Livaja and O.J. Rando. Nucleosome positioning: How is it established, and why does it matter? *Developmental biology*, 339(2):258–266, 2010.

[23] G. Arya, A. Maitra, and SA Grigoryev. A structural perspective on the where, how, why, and what of nucleosome positioning. *Journal of biomolecular structure & dynamics*, 27(6):803, 2010.

[24] R.C. Gentleman, V.J. Carey, D.M. Bates, B. Bolstad, M. Dettling, S. Dudoit, B. Ellis, L. Gautier, Y. Ge, J. Gentry, et al. Bioconductor: open software development for computational biology and bioinformatics. *Genome biology*, 5(10):R80, 2004.

[25] M. Morgan, S. Anders, M. Lawrence, P. Aboyoun, H. Pagès, and R. Gentleman. Shortread: a bioconductor package for input, quality assessment and exploration of high-throughput sequence data. *Bioinformatics*, 25(19):2607, 2009.

[26] G.A. Maston, S.K. Evans, and M.R. Green. Transcriptional regulatory elements in the human genome. *Annu. Rev. Genomics Hum. Genet.*, 7:29–59, 2006.

[27] Z. Wang, C. Zang, K. Cui, D.E. Schones, A. Barski, W. Peng, and K. Zhao. Genome-wide mapping of hats and hdacs reveals distinct functions in active and inactive genes. *Cell*, 138(5):1019–1031, 2009.

[28] A. Rada-Iglesias, R. Bajpai, T. Swigut, S.A. Brugmann, R.A. Flynn, and J. Wysocka. A unique chromatin signature uncovers early developmental enhancers in humans. *Nature*, 470(7333):279–283, 2010.

[29] M.P. Creyghton, A.W. Cheng, G.G. Welstead, T. Kooistra, B.W. Carey, E.J. Steine, J. Hanna, M.A. Lodato, G.M. Frampton, P.A. Sharp, et al. Histone h3k27ac separates active from poised enhancers and predicts developmental state. *Proceedings of the National Academy of Sciences*, 107(50):21931, 2010.

[30] E.D. Rubio, D.J. Reiss, P.L. Welcsh, C.M. Disteche, G.N. Filippova, N.S. Baliga, R. Aebersold, J.A. Ranish, and A. Krumm. Ctcf physically links cohesin to chromatin. *Proceedings of the National Academy of Sciences*, 105(24):8309, 2008.

[31] T.H. Kim, Z.K. Abdullaev, A.D. Smith, K.A. Ching, D.I. Loukinov, R.D. Green, M.Q. Zhang, V.V. Lobanenkov, and B. Ren. Analysis of the vertebrate insulator protein ctcf-binding sites in the human genome. *Cell*, 128(6):1231–1245, 2007.

[32] X. Xie, T.S. Mikkelsen, A. Gnirke, K. Lindblad-Toh, M. Kellis, and E.S. Lander. Systematic discovery of regulatory motifs in conserved regions of the human genome, including thousands of ctcf insulator sites. *Proceedings of the National Academy of Sciences*, 104(17):7145, 2007.

[33] H. Lähdesmäki, I. Shmulevich, and O. Yli-Harja. On learning gene regulatory networks under the boolean network model. *Machine Learning*, 52(1):147–167, 2003.

[34] I. Shmulevich, O. Yli-Harja, J. Astola, et al. Inference of genetic regulatory networks under the best-fit extension paradigm, 2002.

[35] E. Boros, T. Ibaraki, and K. Makino. Error-free and best-fit extensions of partially defined boolean functions* 1. *Information and Computation*, 140(2):254–283, 1998.

[36] P. Pudil, J. Novovicová, and J. Kittler. Floating search methods in feature selection. *Pattern recognition letters*, 15(11):1119–1125, 1994.

[37] I. Shmulevich and S.A. Kauffman. Activities and sensitivities in boolean network models. *Physical Review Letters*, 93(4):48701, 2004.

[38] I. Shmulevich and E.R. Dougherty. *Probabilistic Boolean Networks: The Modeling and Control of Gene Regulatory Networks*. Siam, 2009.

[39] E.L. Greer, T.J. Maures, A.G. Hauswirth, E.M. Green, D.S. Leeman, G.S. Maro, S. Han, M.R. Banko, O. Gozani, and A. Brunet. Members of the h3k4 trimethylation complex regulate lifespan in a germline-dependent manner in c. elegans. *Nature*, 466(7304):383–387, 2010.

[40] E.L. Greer, T.J. Maures, D. Ucar, A.G. Hauswirth, E. Mancini, J.P. Lim, B.A. Benayoun, Y. Shi, and A. Brunet. Transgenerational epigenetic inheritance of longevity in caenorhabditis elegans. *Nature*, 2011.